

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386101
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Feng; Jiangwei et al.

Textured region of a substrate to reduce specular reflectance incorporating surface features with an elliptical perimeter or segments thereof, and method of making the same

Abstract

A substrate for a display article includes: a primary surface; a textured region on at least a portion of the primary surface, the textured region comprising surface features that reflect a random distribution, each of the surface features comprising a perimeter that is parallel to a base-plane extending through a thickness of the substrate below the textured region, wherein the perimeter is elliptical. The textured region can further include (i) one or more higher surfaces residing at a higher mean elevation from the base-plane and (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation. The higher mean elevation can differ from the lower mean elevation by a distance within a range of 0.05 μm to 0.70 μm .

Inventors: Feng; Jiangwei (Painted Post, NY), Isaac; Corinne Elizabeth (Horseheads, NY), Koch, III; Karl William (Elmira, NY), Li; Shenping (Painted Post, NY), Senaratne; Wageesha (Horseheads, NY), Wood; William Allen (Painted Post, NY)

Applicant: Corning Incorporated (Corning, NY)

Family ID: 77155899

Assignee: CORNING INCORPORATED (Corning, NY)

Appl. No.: 17/369315

Filed: July 07, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220011478 A1	Jan. 13, 2022

Related U.S. Application Data

Publication Classification

Int. Cl.: **G02B5/02** (20060101); **C03C15/00** (20060101); **C23C14/02** (20060101); **C23C14/04** (20060101); **C23C16/02** (20060101); **C23C16/04** (20060101); **G02B1/11** (20150101); **G02B1/118** (20150101); **G02B1/14** (20150101); **G02B27/42** (20060101)

U.S. Cl.:

CPC **G02B1/11** (20130101); **C03C15/00** (20130101); **C23C14/021** (20130101); **C23C14/042** (20130101); **C23C16/0227** (20130101); **C23C16/042** (20130101); **G02B1/118** (20130101); **G02B1/14** (20150115); **G02B5/0221** (20130101); **G02B5/0268** (20130101); **G02B27/4205** (20130101); **G02B27/4272** (20130101); **G02B2207/101** (20130101)

Field of Classification Search

CPC: G02B (1/11); G02B (1/118); G02B (1/14); G02B (5/0221); G02B (5/0268); G02B (27/4205); G02B (27/4272); G02B (2207/101); C03C (15/00); C03C (17/3435); C03C (3/076); C03C (3/083); C03C (2204/08); C03C (2218/34); C03C (21/002); C23C (14/021); C23C (14/042); C23C (16/0227); C23C (16/042)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3067021	12/1961	Pelley et al.	N/A	N/A
3150032	12/1963	Rubenstein	N/A	N/A
3413058	12/1967	Fang et al.	N/A	N/A
4579765	12/1985	Schachtner	N/A	N/A
4687707	12/1986	Matsuo et al.	N/A	N/A
4797316	12/1988	Hecq et al.	N/A	N/A
4946923	12/1989	Nagata et al.	N/A	N/A
5470606	12/1994	De Boer	N/A	N/A
5618619	12/1996	Petrmichl et al.	N/A	N/A
5681636	12/1996	Marshall	N/A	G11B 5/73919
5737472	12/1997	Bernasson et al.	N/A	N/A
5855717	12/1998	Beer	264/293	B44F 1/00
6046855	12/1999	Goto	N/A	N/A
6208389	12/2000	Aben et al.	N/A	N/A
6219121	12/2000	Sahouani et al.	N/A	N/A
RE37183	12/2000	Kawamura et al.	N/A	N/A
6254913	12/2000	Wadsworth et al.	N/A	N/A
6286965	12/2000	Caskey et al.	N/A	N/A
6340404	12/2001	Oka et al.	N/A	N/A
6521677	12/2002	Yashiro et al.	N/A	N/A
6723423	12/2003	Kaneko et al.	N/A	N/A

6824709	12/2003	Shundo	N/A	N/A
6846599	12/2004	Ide	N/A	N/A
6862139	12/2004	Chang et al.	N/A	N/A
7037573	12/2005	Miyatake et al.	N/A	N/A
7122253	12/2005	Yamaguchi et al.	N/A	N/A
7128428	12/2005	Takahashi et al.	N/A	N/A
7149032	12/2005	Ohishi et al.	N/A	N/A
7171676	12/2006	Takeda et al.	N/A	N/A
7253861	12/2006	Niiyama et al.	N/A	N/A
7264866	12/2006	Hashimoto et al.	N/A	N/A
7332213	12/2007	Mimura et al.	N/A	N/A
7371439	12/2007	Matsunaga et al.	N/A	N/A
7371786	12/2007	Yoshihara et al.	N/A	N/A
7390099	12/2007	Takao et al.	N/A	N/A
7410686	12/2007	Osada et al.	N/A	N/A
7542207	12/2008	Matsunaga	N/A	N/A
7604358	12/2008	Ninomiya et al.	N/A	N/A
7629400	12/2008	Hyman	N/A	N/A
7645502	12/2009	Mikami et al.	N/A	N/A
7737633	12/2009	Zheng	N/A	N/A
7796123	12/2009	Irvin et al.	N/A	N/A
7799732	12/2009	Tanaka et al.	N/A	N/A
7903340	12/2010	Nagahama et al.	N/A	N/A
7973892	12/2010	Lim	N/A	N/A
8026021	12/2010	Stumpe et al.	N/A	N/A
8062731	12/2010	Takada et al.	N/A	N/A
8110278	12/2011	Hsu et al.	N/A	N/A
8124215	12/2011	Takao et al.	N/A	N/A
8304055	12/2011	Haga et al.	N/A	N/A
8312739	12/2011	Lee et al.	N/A	N/A
8325418	12/2011	Nagahama et al.	N/A	N/A
8514351	12/2012	Sasaki et al.	N/A	N/A
8561429	12/2012	Allan et al.	N/A	N/A
8628896	12/2013	Ikeda et al.	N/A	N/A
8845172	12/2013	Jang et al.	N/A	N/A
8854623	12/2013	Fontaine et al.	N/A	N/A
8888965	12/2013	Kuppuswamy et al.	N/A	N/A
9023457	12/2014	Carrilero et al.	N/A	N/A
9051404	12/2014	Jiang et al.	N/A	N/A
9051423	12/2014	Qiu	N/A	N/A
9079802	12/2014	Bellman et al.	N/A	N/A
9102131	12/2014	Derks et al.	N/A	N/A
9158044	12/2014	Akiyama et al.	N/A	N/A
9263202	12/2015	Lee et al.	N/A	N/A
9316885	12/2015	Lo et al.	N/A	N/A
9335444	12/2015	Hart et al.	N/A	N/A
9359261	12/2015	Bellman et al.	N/A	N/A
9366784	12/2015	Bellman et al.	N/A	N/A
9400420	12/2015	Pudleiner et al.	N/A	N/A

9411180	12/2015	Gollier et al.	N/A	N/A
9418193	12/2015	Dowski et al.	N/A	N/A
9535280	12/2016	Borrelli et al.	N/A	N/A
9550161	12/2016	Arfsten et al.	N/A	N/A
9573842	12/2016	Gollier et al.	N/A	N/A
9581731	12/2016	Bookbinder et al.	N/A	N/A
9588263	12/2016	Gollier et al.	N/A	N/A
9651720	12/2016	Lander et al.	N/A	N/A
9701248	12/2016	Neuman et al.	N/A	N/A
9701579	12/2016	Gollier et al.	N/A	N/A
9703011	12/2016	Adib et al.	N/A	N/A
9718249	12/2016	Kwong	N/A	N/A
9726786	12/2016	Hart et al.	N/A	N/A
9766376	12/2016	Ho et al.	N/A	N/A
9786194	12/2016	Hyman	N/A	N/A
9823209	12/2016	Yu et al.	N/A	N/A
9880328	12/2017	Gollier et al.	N/A	N/A
9896596	12/2017	Jung et al.	N/A	N/A
9939557	12/2017	David et al.	N/A	N/A
9964773	12/2017	Wang	N/A	N/A
9987820	12/2017	Mehlmann et al.	N/A	N/A
10183889	12/2018	Gollier et al.	N/A	N/A
10613340	12/2019	Borrelli et al.	N/A	N/A
10698151	12/2019	Hamilton et al.	N/A	N/A
10725230	12/2019	Jalava	N/A	G02B 6/0068
10899661	12/2020	Gollier et al.	N/A	N/A
10921492	12/2020	Gregorski et al.	N/A	N/A
10948629	12/2020	Hart et al.	N/A	N/A
2002/0085284	12/2001	Nakamura et al.	N/A	N/A
2002/0090507	12/2001	Barth et al.	N/A	N/A
2002/0167629	12/2001	Blanchard	N/A	N/A
2003/0234460	12/2002	Hayashi et al.	N/A	N/A
2004/0005482	12/2003	Kobayashi et al.	N/A	N/A
2004/0184765	12/2003	Difrancesco et al.	N/A	N/A
2004/0188874	12/2003	Hikita et al.	N/A	N/A
2004/0195960	12/2003	Czeremuszkin et al.	N/A	N/A
2004/0233174	12/2003	Robrecht et al.	N/A	N/A
2005/0007019	12/2004	Kim et al.	N/A	N/A
2005/0074591	12/2004	Zagdoun	N/A	N/A
2005/0287309	12/2004	Veerasamy	N/A	N/A
2006/0152801	12/2005	Matsunaga	N/A	N/A
2006/0153979	12/2005	Asakura et al.	N/A	N/A
2006/0240266	12/2005	Schicht et al.	N/A	N/A
2006/0274047	12/2005	Spath et al.	N/A	N/A
2006/0274048	12/2005	Spath et al.	N/A	N/A
2006/0286465	12/2005	Kim	N/A	N/A
2007/0014981	12/2006	Chiang et al.	N/A	N/A
2007/0141357	12/2006	Bekiarian et al.	N/A	N/A

2007/0152985	12/2006	Ostergaard et al.	N/A	N/A
2007/0166536	12/2006	Dollase et al.	N/A	N/A
2007/0240804	12/2006	Arai et al.	N/A	N/A
2007/0249789	12/2006	Buehler et al.	N/A	N/A
2007/0266896	12/2006	Suwa et al.	N/A	N/A
2008/0138606	12/2007	Yoshihara et al.	N/A	N/A
2008/0191463	12/2007	Vermeulen et al.	N/A	N/A
2009/0004462	12/2008	Zhang et al.	N/A	N/A
2009/0017314	12/2008	Nadaud et al.	N/A	N/A
2009/0051668	12/2008	Cheng	N/A	N/A
2009/0104385	12/2008	Reymond et al.	N/A	N/A
2009/0135492	12/2008	Kusuda et al.	N/A	N/A
2009/0178704	12/2008	Kalejs et al.	N/A	N/A
2009/0268299	12/2008	Furui et al.	N/A	N/A
2010/0009154	12/2009	Allan et al.	N/A	N/A
2010/0027383	12/2009	Suzuki et al.	N/A	N/A
2010/0062217	12/2009	Kurematsu	N/A	N/A
2010/0130348	12/2009	Kang et al.	N/A	N/A
2010/0149483	12/2009	Chiavetta, III	N/A	N/A
2010/0167019	12/2009	Ohyanagi et al.	N/A	N/A
2010/0177398	12/2009	Watanabe	216/11	G02B 5/0221
2010/0182551	12/2009	Tochigi et al.	N/A	N/A
2010/0195311	12/2009	Furui et al.	N/A	N/A
2010/0196650	12/2009	Okawa et al.	N/A	N/A
2010/0196685	12/2009	Murata et al.	N/A	N/A
2010/0238384	12/2009	Tochigi et al.	N/A	N/A
2010/0258752	12/2009	Mochizuki et al.	N/A	N/A
2010/0272990	12/2009	Bondesan et al.	N/A	N/A
2010/0316861	12/2009	Kubler et al.	N/A	N/A
2011/0064943	12/2010	Wang	N/A	N/A
2011/0100424	12/2010	Roche et al.	N/A	N/A
2011/0128664	12/2010	Coue et al.	N/A	N/A
2012/0013983	12/2011	Chang et al.	N/A	N/A
2012/0070603	12/2011	Hsu	N/A	N/A
2012/0113043	12/2011	Liu et al.	N/A	N/A
2012/0250135	12/2011	Yeh et al.	N/A	N/A
2012/0270041	12/2011	Matsumoto et al.	N/A	N/A
2012/0291840	12/2011	Kohnke	428/141	C03C 19/00
2012/0300307	12/2011	Borrelli	359/609	G02B 5/0278
2013/0107370	12/2012	Lander	359/609	C03C 3/087
2013/0127202	12/2012	Hart	N/A	N/A
2013/0183489	12/2012	Cremer et al.	N/A	N/A
2014/0090864	12/2013	Paulson	N/A	N/A
2014/0093711	12/2013	Paulson	N/A	N/A
2014/0106141	12/2013	Bellman et al.	N/A	N/A
2014/0113083	12/2013	Lee et al.	N/A	N/A
2014/0131091	12/2013	Smith	N/A	N/A
2014/0154661	12/2013	Bookbinder et al.	N/A	N/A

2014/0268349	12/2013	Kalyankar	427/164	G02B 5/0294
2014/0313441	12/2013	Lim	N/A	N/A
2014/0320422	12/2013	Williams et al.	N/A	N/A
2014/0376094	12/2013	Bellman et al.	N/A	N/A
2015/0079368	12/2014	Koike et al.	N/A	N/A
2015/0111725	12/2014	Van et al.	N/A	N/A
2015/0160376	12/2014	Kohli et al.	N/A	N/A
2015/0174625	12/2014	Hart et al.	N/A	N/A
2015/0177778	12/2014	Chen et al.	N/A	N/A
2015/0185554	12/2014	Zhao et al.	N/A	N/A
2015/0192897	12/2014	Schilling	359/2	B42D 25/355
2015/0250237	12/2014	Shoham et al.	N/A	N/A
2015/0253467	12/2014	Sano	N/A	N/A
2015/0309628	12/2014	Chen et al.	N/A	N/A
2015/0316442	12/2014	Tamada et al.	N/A	N/A
2015/0322270	12/2014	Amin et al.	N/A	N/A
2015/0323705	12/2014	Hart et al.	N/A	N/A
2015/0323812	12/2014	Ishak et al.	N/A	N/A
2015/0355382	12/2014	Henn et al.	N/A	N/A
2016/0002498	12/2015	Maghsoodi et al.	N/A	N/A
2016/0016845	12/2015	Cho et al.	N/A	N/A
2016/0041308	12/2015	Kramer et al.	N/A	N/A
2016/0083835	12/2015	Adib et al.	N/A	N/A
2016/0137548	12/2015	Cabral et al.	N/A	N/A
2016/0137873	12/2015	Kostromine et al.	N/A	N/A
2016/0146978	12/2015	Lee et al.	N/A	N/A
2016/0236974	12/2015	Sinapi et al.	N/A	N/A
2016/0246154	12/2015	O'Keefe	N/A	N/A
2016/0306046	12/2015	Axelsson et al.	N/A	N/A
2016/0362583	12/2015	Naik et al.	N/A	N/A
2017/0003420	12/2016	Berit-Debat et al.	N/A	N/A
2017/0015584	12/2016	Krzyzak et al.	N/A	N/A
2017/0018408	12/2016	Xu	N/A	C04B 37/005
2017/0050349	12/2016	Hara et al.	N/A	N/A
2017/0129806	12/2016	Fujii et al.	N/A	N/A
2017/0183255	12/2016	Walther et al.	N/A	N/A
2017/0199307	12/2016	Hart et al.	N/A	N/A
2017/0210666	12/2016	Chang et al.	N/A	N/A
2017/0235418	12/2016	Inamoto et al.	N/A	N/A
2017/0245962	12/2016	Skamser	N/A	G02B 5/0221
2017/0260620	12/2016	Cheah et al.	N/A	N/A
2017/0276838	12/2016	Oishi et al.	N/A	N/A
2017/0285227	12/2016	Chen et al.	N/A	N/A
2017/0307790	12/2016	Bellman et al.	N/A	N/A
2018/0095303	12/2017	Cho et al.	N/A	N/A
2018/0128957	12/2017	Davis et al.	N/A	N/A

2018/0162091	12/2017	Takeda et al.	N/A	N/A
2018/0203163	12/2017	Thakkar et al.	N/A	N/A
2018/0251398	12/2017	Ikegami et al.	N/A	N/A
2018/0352668	12/2017	Amin et al.	N/A	N/A
2018/0372919	12/2017	Suzuki et al.	N/A	N/A
2019/0039935	12/2018	Couillard et al.	N/A	N/A
2019/0039946	12/2018	Bayne et al.	N/A	N/A
2019/0045038	12/2018	Zhou et al.	N/A	N/A
2019/0062200	12/2018	He et al.	N/A	N/A
2019/0219739	12/2018	Gregorski et al.	N/A	N/A
2020/0057177	12/2019	Hart et al.	N/A	N/A
2020/0133012	12/2019	Sales	N/A	G02B 5/0215
2020/0158917	12/2019	Hart et al.	N/A	N/A
2020/0158922	12/2019	Hayashi et al.	N/A	N/A
2020/0377409	12/2019	Jin et al.	N/A	N/A
2020/0379143	12/2019	Gu et al.	N/A	N/A
2020/0389991	12/2019	Shannon	N/A	H04M 1/0264
2021/0017068	12/2020	Torfs et al.	N/A	N/A
2021/0181382	12/2020	Gregorski et al.	N/A	N/A
2022/0011467	12/2021	Feng et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
199540318	12/1995	AU	N/A
2015252116	12/2014	AU	N/A
2629555	12/2008	CA	N/A
1312450	12/2000	CN	N/A
1318722	12/2000	CN	N/A
1653880	12/2004	CN	N/A
1869736	12/2005	CN	N/A
2859579	12/2006	CN	N/A
1936623	12/2006	CN	N/A
1940601	12/2006	CN	N/A
201165502	12/2007	CN	N/A
201483977	12/2009	CN	N/A
201707457	12/2010	CN	N/A
102109630	12/2010	CN	N/A
201945707	12/2010	CN	N/A
201984393	12/2010	CN	N/A
202171708	12/2011	CN	N/A
202177751	12/2011	CN	N/A
202177765	12/2011	CN	N/A
202182978	12/2011	CN	N/A
102627407	12/2011	CN	N/A
202615053	12/2011	CN	N/A
102923969	12/2012	CN	N/A
103013196	12/2012	CN	N/A

103013219	12/2012	CN	N/A
202904161	12/2012	CN	N/A
103099529	12/2012	CN	N/A
202924088	12/2012	CN	N/A
202924096	12/2012	CN	N/A
103171230	12/2012	CN	N/A
203025361	12/2012	CN	N/A
103254670	12/2012	CN	N/A
103302934	12/2012	CN	N/A
103305816	12/2012	CN	N/A
203260587	12/2012	CN	N/A
203535376	12/2013	CN	N/A
203567294	12/2013	CN	N/A
203620645	12/2013	CN	N/A
103921487	12/2013	CN	N/A
103934756	12/2013	CN	N/A
203689480	12/2013	CN	N/A
103964705	12/2013	CN	N/A
104418511	12/2014	CN	N/A
104559625	12/2014	CN	N/A
104845544	12/2014	CN	N/A
204727835	12/2014	CN	N/A
204894681	12/2014	CN	N/A
105446558	12/2015	CN	N/A
105737103	12/2015	CN	N/A
205368144	12/2015	CN	N/A
105843452	12/2015	CN	N/A
105859148	12/2015	CN	N/A
106113837	12/2015	CN	N/A
205687804	12/2015	CN	N/A
106199812	12/2015	CN	N/A
205818592	12/2015	CN	N/A
106338783	12/2016	CN	N/A
106378880	12/2016	CN	N/A
106431004	12/2016	CN	N/A
106941545	12/2016	CN	N/A
107042642	12/2016	CN	N/A
107076878	12/2016	CN	N/A
107174867	12/2016	CN	N/A
107310209	12/2016	CN	N/A
215365506	12/2020	CN	N/A
112919819	12/2021	CN	N/A
0698798	12/1995	EP	N/A
1069088	12/2000	EP	N/A
1275623	12/2002	EP	N/A
1432874	12/2003	EP	N/A
2149540	12/2009	EP	N/A
3178796	12/2016	EP	N/A
3822234	12/2020	EP	N/A
2966934	12/2011	FR	N/A

2485522	12/2011	GB	N/A
11085	12/2010	ID	N/A
201201777	12/2014	IN	N/A
58-127463	12/1982	JP	N/A
61-019888	12/1985	JP	N/A
02-077434	12/1989	JP	N/A
02-156448	12/1989	JP	N/A
07-290652	12/1994	JP	N/A
07-331115	12/1994	JP	N/A
2000-121806	12/1999	JP	N/A
2000-275404	12/1999	JP	N/A
2001-281402	12/2000	JP	N/A
2001-281406	12/2000	JP	N/A
2001-311806	12/2000	JP	N/A
2002-082207	12/2001	JP	N/A
2002-210906	12/2001	JP	N/A
2002-212317	12/2001	JP	N/A
2003-026826	12/2002	JP	N/A
2003-082127	12/2002	JP	N/A
2004-069878	12/2003	JP	N/A
2004-244594	12/2003	JP	N/A
2004-291303	12/2003	JP	N/A
2004-333901	12/2003	JP	N/A
2005-042072	12/2004	JP	N/A
2005-070724	12/2004	JP	N/A
2005-187639	12/2004	JP	N/A
2005-187640	12/2004	JP	N/A
2005-227415	12/2004	JP	N/A
2005-246296	12/2004	JP	N/A
2005-300576	12/2004	JP	N/A
2006-047504	12/2005	JP	N/A
2006-110754	12/2005	JP	N/A
2006-317957	12/2005	JP	N/A
2006-352105	12/2005	JP	N/A
2007-055064	12/2006	JP	N/A
2007-072372	12/2006	JP	N/A
2007-086521	12/2006	JP	N/A
2007-114377	12/2006	JP	N/A
2007-156205	12/2006	JP	N/A
2007-240707	12/2006	JP	N/A
2007-271953	12/2006	JP	N/A
2007-298667	12/2006	JP	N/A
2008-003425	12/2007	JP	N/A
2008-116596	12/2007	JP	N/A
2008-158156	12/2007	JP	N/A
2009-025384	12/2008	JP	N/A
2009-149468	12/2008	JP	N/A
2009-175725	12/2008	JP	N/A
2010-061044	12/2009	JP	N/A
2010-125719	12/2009	JP	N/A

2010-167410	12/2009	JP	N/A
2011-246365	12/2010	JP	N/A
2012-132022	12/2011	JP	N/A
2012-228811	12/2011	JP	N/A
2012-242837	12/2011	JP	N/A
2013-070093	12/2012	JP	N/A
2013-226666	12/2012	JP	N/A
2013-234571	12/2012	JP	N/A
2015-006650	12/2014	JP	N/A
5650347	12/2014	JP	N/A
5736214	12/2014	JP	N/A
2015-167470	12/2014	JP	N/A
2015-171770	12/2014	JP	N/A
2016-009172	12/2015	JP	N/A
2016-201236	12/2015	JP	N/A
10-2005-0043647	12/2004	KR	N/A
10-2006-0024545	12/2005	KR	N/A
10-2006-0060171	12/2005	KR	N/A
10-2006-0065724	12/2005	KR	N/A
10-0709879	12/2006	KR	N/A
10-2007-0054850	12/2006	KR	N/A
10-2007-0063134	12/2006	KR	N/A
10-2008-0048578	12/2007	KR	N/A
10-2008-0057443	12/2007	KR	N/A
10-2009-0098975	12/2008	KR	N/A
10-2009-0119968	12/2008	KR	N/A
10-2010-0013836	12/2009	KR	N/A
10-2010-0123624	12/2009	KR	N/A
2011-0047596	12/2010	KR	N/A
10-2011-0078682	12/2010	KR	N/A
10-1121207	12/2011	KR	N/A
2013-0031689	12/2012	KR	N/A
10-2014-0061842	12/2013	KR	N/A
10-1517051	12/2014	KR	N/A
10-2015-0116802	12/2014	KR	N/A
10-2017-0028190	12/2016	KR	N/A
10-2017-0043566	12/2016	KR	N/A
02/00772	12/2001	WO	N/A
03/09767	12/2002	WO	N/A
03/27397	12/2002	WO	N/A
2007/049589	12/2006	WO	N/A
2008/062605	12/2007	WO	N/A
2009/008240	12/2008	WO	N/A
2009/037886	12/2008	WO	N/A
2009/065490	12/2008	WO	N/A
2009/107536	12/2008	WO	N/A
2010/114135	12/2009	WO	N/A
2013/023359	12/2012	WO	N/A
2014/061614	12/2013	WO	N/A
2014/085414	12/2013	WO	N/A

2014/117333	12/2013	WO	N/A
2015/015338	12/2014	WO	N/A
2015/030118	12/2014	WO	N/A
2015/084253	12/2014	WO	N/A
2015/095288	12/2014	WO	N/A
2015/108266	12/2014	WO	N/A
2015/115154	12/2014	WO	N/A
2015/137196	12/2014	WO	N/A
2015/174625	12/2014	WO	N/A
2015/190374	12/2014	WO	N/A
2016/069113	12/2015	WO	N/A
2016/204009	12/2015	WO	N/A
2016/205317	12/2015	WO	N/A
2017/041307	12/2016	WO	N/A
2018/043253	12/2017	WO	N/A
2018/125676	12/2017	WO	N/A
2019/055745	12/2018	WO	N/A
2020/013012	12/2019	WO	N/A
2020/123367	12/2019	WO	N/A
2022/011070	12/2021	WO	N/A
2022/011071	12/2021	WO	N/A
2022/125846	12/2021	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion of the International Searching Authority; PCT/US2021/040773; dated Oct. 26, 2021; 14 pages; European Patent Office. cited by applicant

“Cieluv”, Available at: <https://en.wikipedia.org/wiki/CIELUV>, 1976, 5 pages. cited by applicant

“Heron's formula”, Available at: https://en.wikipedia.org/wiki/Heron%27s_formula, 8 pages. cited by applicant

“High-precision, fast array spectroradiometer—for demanding measurements”, Available at: <https://www.instrumentsystems.com/en/products/spectrometers/cas-140d>, 4 pages. cited by applicant

“Illuminance Meters”, Available at: <https://sensing.konicaminolta.us/us/technologies/illuminance-meters/>, 2 pages. cited by applicant

“Welcome to Display—Metrology & Systems”, Available at: www.display-messtechnik.de/en/measurement, 2014, 1 page. cited by applicant

Al-Dahoudi, N. et al., “Transparent conducting, anti-static and anti-static-anti-glare coatings on plastic substrates,” Thin Solid Films 2001, 392, pp. 299-304. cited by applicant

ASTM C770-16(2020), “Standard Test Method for Measurement of Glass Stress-Optical Coefficient,” ASTM International, West Conshohocken, PA, 2020, www.astm.org. cited by applicant

ASTM D1003-21, “Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics,” ASTM International, West Conshohocken, PA, 2021, www.astm.org. cited by applicant

ASTM D523-14(2018), “Standard Test Method for Specular Gloss,” ASTM International, West Conshohocken, PA, 2018, www.astm.org. cited by applicant

ASTM E2001-13, “Standard Guide for Resonant Ultrasound Spectroscopy for Defect Detection in Both Metallic and Non-metallic Parts,” ASTM International, West Conshohocken, PA, 2013, www.astm.org. cited by applicant

Cecala, C. et al., “Fourier Optics Modeling of Display Sparkle from Anti-Glare Cover Glass: Comparison to Experimental Data,” Imaging and Applied Optics Congress, OSA Technical Digest

(Optical Society of America, 2020), paper JW5B.8, 2 pages. cited by applicant

Chou, S.Y. et al., "Imprint of Sub-25 nm Vias and Trenches in Polymers," Appl. Phys. Lett. 67, 3114, 1995, 3 pages. cited by applicant

Cook, R. L., "Stochastic sampling in computer graphics," Computer Graphics (Proceedings of ACM SIGGRAPH 86) 5, 1 (1986), pp. 51-72. cited by applicant

Kelley, E., "Proposed Diffuse Ambient Contrast Measurement Methods for Flat Panel Displays," 2001, NISTIR 6738, National Institute of Standards and Technology, Gaithersburg, Maryland, 7 pages. cited by applicant

Kittel, C., Introduction to Solid State Physics, seventh edition, John Wiley & Sons, Inc., NY, 1996, pp. 611-627. cited by applicant

Liu, B.T. et al., "Strength of the interactions between light-scattering particles and resins affects the haze of anti-glare films," Colloids Surf. a Physicochem. Eng. Aspects 2011, 389, pp. 138-143. cited by applicant

Modified ASTM D5767-18 test protocol, "Standard Test Method for Instrumental Measurement of Distinctness-of-Image (DOI) Gloss of Coated Surfaces using a Rhopoint IQ Gloss Haze & DOI Meter," Rhopoint Instruments Ltd. cited by applicant

Schlömer, T. et al., "Farthest-points optimized point sets with maximized minimum distance," Proc. ACM SIGGRAPH Symposium on high performance graphics: HPG 2011 pp. 135-142. cited by applicant

Shackelford, J.F., Introduction to Materials Science for Engineers, Sixth Edition, Pearson Prentice Hall, New Jersey, 2005, pp. 404-418. cited by applicant

Simonsen, et al., "Haze of random systems: an approximate analytic approach," Phys Rev A 79 063813 (2009), 23 pages. cited by applicant

Smart and Moore, Solid State Chemistry, an introduction, Chapman & Hall University and Professional Division, London, 1992, pp. 136-151. cited by applicant

Stillwell, A. et al., "Perception of Sparkle in Anti-Glare Display Screens," JSID 22(2), 129-136 (2014). cited by applicant

Boerner et al; "Holographic Antiglare and Antireflection Films for Flat Panel Displays" SID 03 DIGEST, 7.3, p. 68, 2003. cited by applicant

International Search Report and Written Opinion of the International Searching Authority; PCT/US2019/012807; Mailed Mar. 29, 2019; 12 Pages; European Patent Office. cited by applicant

Oliver et al; "An Improved Technique for Determining Hardness and Elastic Modulus Using Load and Displacement Sensing Indentation Experiments"; J. Mater. Res., vol. 7, No. 6, 1992, pp. 1564-1583. cited by applicant

Oliver et al; "Measurement of Hardness and Elastic Modulus by Instrument Indentation: Advances in Understanding and Refinements to Methodology"; J. Mater. Res., vol. 19, No. 1, 2004, pp. 3-20. cited by applicant

Taguchi et al; "Ultra-Low-Reflective 60-In. LCD With Uniform Moth-Eye Surface for Digital Signage," SID 10 DIGEST, 80.3, p. 1196, 2010. cited by applicant

International Search Report and Written Opinion of the International Searching Authority; PCT/US2021/040772; dated Nov. 10, 2021; 10 pages; European Patent Office. cited by applicant

ASTM Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics; Designation D 1003-07, Nov. 2007. cited by applicant

International Search Report and Written Opinion of the International Searching Authority; PCT/US2022/034624; dated Oct. 25, 2022; 11 pages; European Patent Office. cited by applicant

International Search Report and Written Opinion of the International Searching Authority; PCT/US2022/034812; dated Oct. 24, 2022; 12 pages; European Patent Office. cited by applicant

Radhakrishnan et al., "Order parameters in molecular simulations", Handbook of Materials Modeling, Chapter 6, 2005, pp. 1613-1626. cited by applicant

Primary Examiner: Stoffa; Wyatt A

Assistant Examiner: Sipes; John Curtis

Background/Summary

CLAIM OF PRIORITY (1) This patent application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/049,843, entitled “DISPLAY ARTICLES WITH DIFFRACTIVE, ANTIGLARE SURFACES AND METHODS OF MAKING THE SAME,” filed Jul. 9, 2020, the entire disclosure of which is incorporated herein by reference.

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application relates to, but does not claim priority to, commonly owned and assigned U.S. Pat. No. 12,147,009, entitled “TEXTURED REGION TO REDUCE SPECULAR REFLECTANCE INCLUDING A LOW REFRACTIVE INDEX SUBSTRATE WITH HIGHER ELEVATED SURFACES AND LOWER ELEVATED SURFACES AND A HIGH REFRACTIVE INDEX MATERIAL DISPOSED ON THE LOWER ELEVATED SURFACES” and filed on Jul. 7, 2021; U.S. patent application Ser. No. 17/369,301, entitled “ANTI-GLARE SUBSTRATE FOR A DISPLAY ARTICLE INCLUDING A TEXTURED REGION WITH PRIMARY SURFACE FEATURES AND SECONDARY SURFACE FEATURES IMPARTING A SURFACE ROUGHNESS THAT INCREASES SURFACE SCATTERING” and filed on Jul. 7, 2021; U.S. Pat. No. 11,977,206, entitled “DISPLAY ARTICLES WITH DIFFRACTIVE, ANTIGLARE SURFACES AND THIN, DURABLE ANTIREFLECTION COATINGS” and filed on Jul. 8, 2021; and U.S. Pat. No. 11,971,519, entitled “DISPLAY ARTICLES WITH ANTIGLARE SURFACES AND THIN, DURABLE ANTIREFLECTION COATINGS” and filed on Jul. 8, 2021. The entire disclosures of each of the foregoing U.S. patent applications, publications and patent documents are incorporated herein by reference.

BACKGROUND

(2) Substrates transparent to visible light are utilized to cover displays of display articles. Such display articles include smart phones, tablets, televisions, computer monitors, and the like. The displays are often liquid crystal displays, organic light emitting diodes, among others. The substrate protects the display, while the transparency of the substrate allows the user of the device to view the display.

(3) The substrate reflecting ambient light, especially specular reflection, reduces the ability of the user to view the display through the substrate. Specular reflection in this context is the mirror-like reflection of ambient light off the substrate. For example, the substrate may reflect visible light reflecting off or emitted by an object in the environment around the device. The visible light reflecting off the substrate reduces the contrast of the light from the display transmitting to the eyes of the user through the substrate. At some viewing angles, instead of seeing the visible light that the display emits, the user sees a specularly reflected image. Thus, attempts have been made to reduce specular reflection of visible ambient light off the substrate.

(4) Attempts have been made to reduce specular reflection off the substrate by texturing the reflecting surface of the substrate. The resulting surface is sometimes referred to as an “antiglare surface.” For examples, sandblasting and liquid etching the surface of the substrate can texture the surface, which generally causes the surface to reflect ambient light diffusely rather than specularly. Diffuse reflection generally means that the surface still reflects the same intensity of ambient light

but the texture of the reflecting surface scatters the light upon reflection. The more diffuse reflection interferes less with the ability of the user to see the visible light that the display emits.

(5) Such methods of texturing (i.e., sandblasting and liquid etching) generate features on the surface with imprecise and unrepeatable geometry (the features provide the texture). The geometry of the textured surface of one substrate formed via sandblasting or liquid etching can never be exactly the same as the geometry of the textured surface of another substrate formed via sandblasting or liquid etching. Commonly, only a quantification of the surface roughness (i.e., R_a) of the textured surface of the substrate is a repeatable target of the texturing.

(6) There are a variety of metrics by which the quality of the “antiglare” surface is judged. Those metrics include (1) the distinctness-of-image, (2) pixel power deviation, (3) apparent Moiré interference fringes, (4) transmission haze, and (5) reflection color artifacts. Distinctness-of-image, which more aptly might be referred to as distinctness-of-reflected-image, is a measure of how distinct an image reflecting off the surface appears. The lower the distinctness-of-image, the more the textured surface is diffusely reflecting rather than specularly reflecting. Surface features can magnify various pixels of the display, which distorts the image that the user views. Pixel power deviation, also referred to as “sparkle,” is a quantification of such an effect. The lower the pixel power deviation the better. Moiré interference fringes are large scale interference patterns, which, if visible, distort the image that the user sees. Preferably, the textured surface produces no apparent Moiré interference fringes. Transmission haze is a measure of how much the textured surface is diffusing the visible light that the display emitted upon transmitting through the substrate. The greater the transmission haze, the less sharp the display appears (i.e., lowered apparent resolution). Reflection color artifacts are a sort of chromatic aberration where the textured surface diffracts light upon reflection as a function of wavelength—meaning that the reflected light, although relatively diffuse, appears segmented by color. The less reflected color artifacts that the textured surface produces the better. All of these attributes are discussed in greater detail below.

(7) Targeting a specific surface roughness cannot optimize all of those metrics simultaneously. A relatively high surface roughness that sandblasting or liquid etching produces might adequately transform specular reflection into diffuse reflection. However, the high surface roughness can additionally generate high transmission haze and pixel power deviation. A relatively low surface roughness, while decreasing transmission haze, might fail to sufficiently transform specular reflection into diffuse reflection—defeating the “antiglare” purpose of the texturing.

(8) Accordingly, a new approach to providing a textured region of the substrate is needed—one that causes the textured surface to reflect ambient light sufficiently diffusely rather than specularly so as to be “antiglare” (e.g., a low distinctness-of-image) but simultaneously also delivers low pixel power deviation, low transmission haze, and low reflection color artifacts.

SUMMARY

(9) The present disclosure addresses that need with a textured region of a substrate that incorporates randomly distributed and oriented elliptical surface features or segments thereof. The surface features having such an elliptical or partially elliptical shape, when randomly distributed and oriented, provide low specular reflection with minimal transmission haze and pixel power deviation and with minimal or unperceivable reflected color artifacts. Although the surface features are randomly distributed and oriented, the surface features are specifically placed by design. Thus, the textured region can be reproduced from substrate to substrate.

(10) According to a first aspect of the present disclosure, a substrate for a display article, the substrate comprising: (i) a primary surface; and (ii) a textured region on at least a portion of the primary surface, the textured region comprising surface features that reflect a random distribution, each of the surface features comprising a perimeter that is parallel to a base-plane extending through a thickness of the substrate below the textured region, wherein the perimeter is elliptical.

(11) According to a second aspect of the present disclosure, the substrate of the first aspect, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher

mean elevation from the base-plane and (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation.

(12) According to a third aspect of the present disclosure, the substrate of the second aspect, wherein (a) the textured region further comprises a surrounding portion, into which the surface features are set, or from which the surface features project; (b) the surrounding portion provides either (i) the one or more higher surfaces or (ii) the one or more lower surfaces; and (c) the surface features provide the other of (i) the one or more higher surfaces and (ii) the one or more lower surfaces, whichever the surrounding portion does not provide.

(13) According to a fourth aspect of the present disclosure, the substrate of any one of the second through third aspects, wherein the higher mean elevation differs from the lower mean elevation by a distance within a range of 0.05 μm to 0.70 μm .

(14) According to a fifth aspect of the present disclosure, the substrate of any one of the first through fourth aspects, wherein (i) the perimeter of each of the surface features comprises a longest dimension parallel to the base-plane; and (ii) the longest dimensions of the perimeters of the surface features are not all parallel to each other.

(15) According to a sixth aspect of the present disclosure, the substrate of the fifth aspect, wherein the longest dimension of the perimeter of each of the surface features is within a range of 5 μm to 150 μm .

(16) According to a seventh aspect of the present disclosure, the substrate of any one of the first through sixth aspects, wherein the surface features comprise a fill-fraction that is within a range of 40% to 60%.

(17) According to an eighth aspect of the present disclosure, the substrate of the first aspect, wherein (i) the surface features comprise larger surface features and smaller surface features; (ii) the perimeters of the larger surface features comprise a range of longest dimensions parallel to the base-plane; (iii) the perimeters of the smaller surface features comprise another range of longest dimensions parallel to the base-plane; and (iv) the longest dimensions of the range of longest dimensions of the larger surface features are longer than the longest dimensions of the range of longest dimensions of the smaller surface features.

(18) According to a ninth aspect of the present disclosure, the substrate of the eighth aspect, wherein (i) the perimeters of the larger surface features totally surround the perimeters of some of the smaller surface features; and (ii) the perimeters of some the smaller surface features reside entirely outside of the perimeters.

(19) According to a tenth aspect of the present disclosure, the substrate of the eighth aspect, wherein the perimeter of at least one of the smaller surface features partially overlaps with the perimeter of one of the larger surface features, such that (i) part of the perimeter of the smaller surface features is inside the perimeter of the larger surface feature and (ii) part of the perimeter of the smaller surface features is outside the perimeter of the larger surface feature.

(20) According to an eleventh aspect of the present disclosure, the substrate of any one of the eighth through tenth aspects, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation, and (iii) a surrounding portion, into which the larger surface features are set, or from which the larger surface features project.

(21) According to a twelfth aspect of the present disclosure, the substrate of the eleventh aspect, wherein (i) the larger surface features project from the surrounding portion, some of the smaller surface features project from the surrounding portion, and the remaining smaller surface features, those that do not project from the surrounding portion, are set into the larger surface features; (ii) the larger surface features and smaller surface features that project from the surrounding portion provide the one or more higher surfaces residing at the higher mean elevation; and the surrounding portion and the smaller surface features that are set into larger surface features provide the one or

more lower surfaces residing at the lower mean elevation.

(22) According to a thirteenth aspect of the present disclosure, the substrate of the eleventh aspect, wherein (i) the larger surface features are set into the surrounding portion, some of the smaller surface features are set into the surrounding portion, and the remaining smaller surface features, those that are not set into the surrounding portion, project from the larger surface features; (ii) the larger surface features and the smaller surface features that are set into the surrounding portion provide the one or more lower surfaces residing at the lower mean elevation; and (iii) the surrounding portion and the smaller surface features that project from the larger surface features provide the one or more higher surfaces residing at the higher mean elevation.

(23) According to a fourteenth aspect of the present disclosure, the substrate of the eighth aspect, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation, (iii) one or more intermediate surfaces residing at one or two intermediate mean elevations from the base-plane, wherein the one or two intermediate mean elevations are disposed between the higher mean elevation and the lower mean elevation, and (iv) a surrounding portion, into which the larger surface features are set, or from which the larger surface features project.

(24) According to a fifteenth aspect of the present disclosure, the substrate of the fourteenth aspect, wherein the larger surface features provide at least a portion of the one or more higher surfaces residing at the higher mean elevation.

(25) According to a sixteenth aspect of the present disclosure, the substrate of the fourteenth aspect, wherein the larger surface features provide at least a portion of the one or more lower surfaces disposed at the lower mean elevation.

(26) According to a seventeenth aspect of the present disclosure, the substrate of any one of the first through sixteenth aspects, wherein the textured region further comprises one or more sections comprising secondary surface features imparting a surface roughness ($R_{\text{sub.a}}$) within a range of 5 nm to 100 nm.

(27) According to an eighteenth aspect of the present disclosure, the substrate of any one of the first through seventeenth aspects, wherein (i) the textured region exhibits a transmission haze within a range of 0.5% to 5.0%; (ii) the textured region exhibits a pixel power deviation within a range of 1.0% to 3.0%; (iii) the textured region exhibits a distinctness-of-image within a range of 5% to 70%; (iv) the textured region exhibits a specular reflectance within a range of 4 GU to 40 GU; and (v) the textured region exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ that are each respectively within a range of 0.03 to 0.6.

(28) According to a nineteenth aspect of the present disclosure, the substrate of any one of the first through eighteenth aspects, wherein the substrate comprises a glass or glass-ceramic.

(29) According to a twentieth aspect of the present disclosure, a substrate for a display article, the substrate comprising: (a) a primary surface; (b) a base-plane extending through the substrate below the primary surface; and (c) a textured region at the primary surface, the textured region comprising (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane, (iii) a first portion providing either the one or more higher surfaces or the one or more lower surfaces, and (iv) a second portion providing the other of the one or more higher surfaces and the one or more higher surfaces, whichever the first portion is not providing; wherein, elliptical perimeters which lie in planes parallel to the base-plane and reflect a random distribution, define the first portion except for overlapping portions of the elliptical perimeters, where the overlapping portions define in part the second portion.

(30) According to a twenty-first aspect of the present disclosure, the substrate of the twentieth aspect, wherein in addition to the overlapping portions of the elliptical perimeters, the second portion of the textured region further comprises any part of the textured region that is not the first

portion.

(31) According to a twenty-second aspect of the present disclosure, the substrate of any one of the twentieth through twenty-first aspects, wherein (i) each of the elliptical perimeters comprises a longest dimension parallel to the base-plane; and (ii) the longest dimensions of the elliptical perimeters are not all parallel with each other.

(32) According to a twenty-third aspect of the present disclosure, the substrate of any one of the twentieth through twenty-second aspects, wherein the higher mean elevation differs from the second elevation by a distance within a range of 0.02 μm to 0.70 μm .

(33) According to a twenty-fourth aspect of the present disclosure, the substrate of any one of the twentieth through twenty-third aspects, wherein (a) the elliptical perimeters comprise larger elliptical perimeters and smaller elliptical perimeters, which are smaller than the first elliptical perimeters; and (b) the first portion of the textured region is bounded by (i) the smaller elliptical perimeters that do not overlap or intersect with the larger elliptical perimeters, (ii) portions of the smaller elliptical perimeters outside of the first elliptical perimeters that partially overlap with the larger elliptical perimeters, and (iii) portions of the larger elliptical perimeters that do not overlap with the smaller elliptical perimeters.

(34) According to a twenty-fifth aspect of the present disclosure, the substrate of the twenty-fourth aspect, wherein at least some of the larger elliptical perimeters entirely encompass more than one of the smaller elliptical perimeters.

(35) According to a twenty-sixth aspect of the present disclosure, the substrate of any one of the twenty-fourth through twenty-fifth aspects, wherein (i) a fill-fraction of the larger elliptical perimeters is within a range of 40% to 60%; and (ii) a fill-fraction of the smaller elliptical perimeters is within a range of 10% to 30%.

(36) According to a twenty-seventh aspect of the present disclosure, the substrate of any one of the twentieth through twenty-sixth aspects, wherein the textured region further comprises one or more sections comprising secondary surface features imparting a surface roughness ($R_{\text{sub.a}}$) within a range of 5 nm to 100 nm.

(37) According to a twenty-eighth aspect of the present disclosure, the substrate of any one of the twentieth through twenty-seventh aspects, wherein (i) the textured region exhibits a transmission haze within a range of 0.3% to 8.0%; (ii) the textured region exhibits a pixel power deviation within a range of 0.7% to 3.5%; (iii) the textured region exhibits a distinctness-of-image within a range of 25% to 100%; (iv) the textured region exhibits a specular reflectance within a range of 5 GU to 30 GU; and (v) the textured region exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ that are each respectively within a range of 0.00 to 0.50.

(38) According to a twenty-ninth aspect of the present disclosure, the substrate of any one of the twentieth through twenty-eighth aspects, wherein the substrate comprises a glass or glass-ceramic.

(39) According to a thirtieth aspect of the present disclosure, a method of forming a textured region of a substrate of a display article, the method comprising: (I) generating a random distribution of ellipses within an area; (II) preparing a lithography mask comprising (a) an area matching the area of the plane; (b) material throughout the matching area; and (c) voids through the material, wherein the random distribution of the ellipses define either (i) the material throughout the matching area or (ii) the voids through the material; (III) exposing a workpiece comprising a substrate, a lithography ink disposed on the substrate, and the lithography mask disposed on the lithography ink, to a curing agent that transmits through the voids through the material of the lithography mask to cure exposed portions of the lithography ink, wherein non-exposed portions of the lithography ink blocked from exposure to the curing agent are not cured and removed along with the lithography mask, and the exposed portions of the lithography ink remain on the substrate as an etching mask; and (IV) contacting the substrate with the etching mask with an etchant, thus forming the textured region.

(40) According to a thirty-first aspect of the present disclosure, the method of the thirtieth aspect further comprises: (I) generating a random distribution of second ellipses within a second area; (II)

preparing a second lithography mask comprising (a) a second area matching the area of the second plane; (b) material throughout the matching second area; and (c) voids through the material, wherein the random distribution of the second ellipses define either (i) the material throughout the matching second area or (ii) the voids through the material; (III) exposing a second workpiece comprising the substrate, new lithography ink disposed on the textured region of the substrate, and the second lithography mask disposed on the new lithography ink, to a curing agent that transmits through the voids through the material of the second lithography mask to cure exposed portions of the new lithography ink, wherein non-exposed portions of the new lithography ink blocked from exposure to the curing agent are not cured and removed along with the second lithography mask, and the exposed portions of the new lithography ink remain on the substrate as a second etching mask; and (IV) contacting the substrate with the second etching mask with an etchant.

(41) According to a thirty-second aspect of the present disclosure, the method of any one of the thirtieth through thirty-first aspects, wherein generating the random distribution of the ellipses comprises: (i) distributing points randomly within an area; (ii) triangulating the points so that each point is made a vertex of a triangle thus forming a plurality of triangles, and none of the triangles overlap; (iii) drawing an ellipse inside each triangle of the plurality of triangles; and (iv) removing the points and the triangles so that only the ellipses that were drawn in the triangles remain on the area.

(42) According to a thirty-third aspect of the present disclosure, the method of the thirty-second aspect, wherein the points randomly distributed within the area are separated by a minimum distance.

(43) According to a thirty-fourth aspect of the present disclosure, the method of any one of the thirtieth through thirty-third aspects further comprises: forming secondary surface features into one or more sections of the textured region, thereby increasing the surface roughness ($R_{sub.a}$) of at the one or more sections to within a range of 5 nm to 100 nm.

(44) According to a thirty-fifth aspect of the present disclosure, a method of forming a textured region of a substrate of a display article, the method comprises: (I) generating a random distribution of first ellipses within a first area; (II) generating a random distribution of second ellipses within a second area, wherein the first ellipses comprise longest dimensions that are on average longer than longest dimensions of the second ellipses on average; (III) forming a new area that superimposes the first ellipses of the first area and the second ellipses of the second area, the new area comprising (a) free portions of the first ellipses and the second ellipses that do not overlap; (b) overlapping portions of the first ellipses and the second ellipses; and (c) empty portions where neither the first ellipses nor the second ellipses are present; (IV) preparing a lithography mask comprising (a) an area; (b) material within the area defined by either (i) the free portions or (ii) a combination of the overlapping portions and the empty portions; and (c) voids through the material defined by whichever of (i) the free portions and (ii) the combination of the overlapping portions and the empty portions do not define the material within the area; (V) exposing a workpiece comprising a substrate, a lithography ink disposed on the substrate, and the lithography mask disposed on the lithography ink, to a curing agent that transmits through the voids through the material of the lithography mask to cure exposed portions of the lithography ink, wherein non-exposed portions of the lithography ink blocked from exposure to the curing agent are not cured and removed along with the lithography mask, and the exposed portions of the lithography ink remain on the substrate as an etching mask; and (VI) contacting the substrate with the etching mask with an etchant.

(45) According to a thirty-sixth aspect of the present disclosure, the method of the thirty-fifth aspect further comprises: before preparing the lithography mask, removing from the new area any of the second ellipses that the first ellipses partially but not fully overlap.

(46) According to a thirty-seventh aspect of the present disclosure, the method of any one of the thirty-fifth through thirty-sixth aspects, wherein generating the random distribution of the first ellipses and generating the random distribution of the second ellipses both separately comprise: (i)

distributing points randomly within an area, each of the points separated by a minimum distance; (ii) triangulating the points so that each point is made a vertex of a triangle thus forming a plurality of triangles, and none of the triangles overlap; (iii) drawing an ellipse inside each triangle of the plurality of triangles; and removing the points and the triangles so that only the ellipses that were drawn in the triangles remain on the area.

(47) According to a thirty-eighth aspect of the present disclosure, the method of the thirty-seventh aspect, wherein (i) the minimum distance separating each of the points of the area for the random distribution of the first ellipses is within a range of 80 μm to 130 μm ; and (ii) the minimum distance separating each of the points of the area for the random distribution of the second ellipses is within a range of 15 μm to 40 μm .

(48) According to a thirty-ninth aspect of the present disclosure, the method of the thirty-seventh aspect, wherein (i) the minimum distance separating each of the points of the area for the random distribution of the first ellipses is less than or equal 20 μm ; (ii) the minimum distance separating each of the points of the area for the random distribution of the second ellipses is less than or equal 18 μm ; and the minimum distance separating each of the points of the area for the random distribution of the first ellipses differs from the minimum distance separating each of the points of the plane for the random distribution of the second ellipses by a value within a range of 1 μm to 3 μm .

(49) According to a fortieth aspect of the present disclosure, the method of any one of the thirty-fifth through thirty-ninth aspects further comprises: forming secondary surface features into one or more sections of the textured region, thereby increasing the surface roughness (R.sub.a) of at the one or more sections to within a range of 5 nm to 100 nm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the figures:

(2) FIG. 1 is a perspective view of a display article, illustrating a substrate including a primary surface with a textured region;

(3) FIG. 2 is a magnified illustration of a portion of an embodiment of the textured region of FIG. 1, illustrating elliptical surface features projecting from a surrounding portion;

(4) FIG. 3 is an elevation view of a cross-section taken through line III-III of FIG. 2, illustrating the elliptical surface features projecting from the surrounding portion, with the elliptical surface features providing higher surfaces that reside at a higher mean elevation from a base-plane extending through the substrate below the textured region, and with the surrounding portion providing a lower surface that resides at a lower mean elevation from the base-plane;

(5) FIG. 4 is a close-up perspective view of area IV of FIG. 2, illustrating an elliptical surface feature projecting from the surrounding portion and providing a higher surface, which is relatively planar;

(6) FIG. 5 is a close-up view of area V of FIG. 2, illustrating the elliptical surface features having a perimeter and a longest dimension, as well as the surface features being separated by a center-to-center distance;

(7) FIG. 6 is a magnified illustration of a portion of an embodiment of the textured region of FIG. 1, illustrating larger elliptical surface features projecting from or set into a surrounding portion and smaller elliptical surface features projecting from or set into the larger elliptical surface features and projecting from or set into the surrounding portion;

(8) FIG. 7 is a close-up view of area VII of FIG. 6, illustrating the smaller elliptical surface features having a shorter longest dimension than the longest dimension of the larger elliptical surface features;

(9) FIGS. 8A-8C are elevational views of a cross-section of FIG. 6, illustrating several possible variations for relative elevations of the larger elliptical surface features, the smaller elliptical surface features, and the surrounding portion from the base-plane, depending on whether a two-step or one-step etching process was utilized to form the textured region;

(10) FIGS. 9A and 9B are magnified illustrations of a portion of an embodiment of the textured region of FIG. 1, illustrating a first portion at a first elevation from the base-plane and a second portion at a second elevation from the base-plane, both incorporating elliptical surface features or segments thereof, a consequence of superimposing a first random distribution of larger ellipses and a second random distribution of smaller ellipses;

(11) FIGS. 10A and 10B are like FIGS. 9A and 9B but are a consequence of superimposing larger ellipses that are closer in size to the smaller ellipses than at FIGS. 9A and 9B;

(12) FIGS. 10C and 10D are elevational views of a cross-section taken through line XC-XC of FIG. 10B, illustrating the first portion and the second portion residing at different elevations from the base-plane, either with the first portion projecting from the second portion (FIG. 10C) or with the first portion set into the second portion (FIG. 10D);

(13) FIG. 11 is a schematic diagram of a set-up used to measure a color shift of ambient light reflecting off the textured region of FIG. 1;

(14) FIG. 12 is a schematic flow chart of a method of forming an embodiment of the textured region of FIG. 1, illustrating steps of randomly distributing ellipses within an area, forming a lithography mask that incorporates the design of the area as a positive or a negative, forming an etching mask that incorporates the design of the area as a positive or negative on the substrate by using the lithography mask to selectively cure lithography ink, and then etching the substrate with the etching mask to form the textured region as either a positive or a negative of the area with the randomly distributed ellipses;

(15) FIG. 13 is a schematic illustrating an example generation of a random distribution of ellipses, where points are randomly generated (such as by using a Poisson disk random distribution algorithm), the points are triangulated, and then Steiner inellipses are drawn within the triangles, which are then later scaled down to a desired fill fraction;

(16) FIG. 14 is a schematic flow chart of another method of forming an embodiment of the textured region of FIG. 1, illustrating two different areas each including a random distribution of ellipses superimposed into one area, with overlapping portions of the ellipses differentiated from non-overlapping portions;

(17) FIGS. 15A-15C, pertaining to a Comparative Example 1A, illustrate circular surface features producing color-separated reflected color artifacts;

(18) FIGS. 16A-16C, pertaining to an Example 1B, illustrate an embodiment of the textured region of FIG. 1 incorporating elliptical surface features not producing color-separated reflected color artifacts;

(19) FIGS. 17A and 17B, pertaining to an Example 2A and a Comparative Example 2B, illustrate elliptical surface features again producing less reflected color artifacts than circular color features, a consequence of having peak intensity of reflected light at close to 0.3 degrees scattering angle and only a minor secondary peak at 3 degrees scattering angle where the scattered intensity is reduced;

(20) FIG. 18A, pertaining to Examples 14A-14O, is a graph that illustrates that the presence of secondary surface features (to impart surface roughness) results in a lower pixel power deviation compared to substrates that did not have the secondary surface features;

(21) FIG. 18B, pertaining to Examples 14A-14O, is a graph that illustrates that the presence of secondary surface features results in a lower specular reflectance compared to substrates that did not have the secondary surface features;

(22) FIG. 18C, pertaining to Examples 14A-14O, is a graph that illustrates that the presence of secondary surface features resulted in a higher distinctness-of-image compared to substrates that

did not have the secondary surface features;

(23) FIG. **18D**, pertaining to Examples 14A-14O, is a graph that illustrates that the presence of secondary surface features resulted in a higher transmission haze compared to substrates that did not have the secondary surface features;

(24) FIG. **19A**, pertaining to Examples 15A-15B, are atomic force microscopy images of the elliptical surface features and the surrounding portion (left) and the secondary surface features (middle and right), for both when the secondary surface features were disposed only on the elliptical surface features (top) and when the secondary surface features were disposed over both the elliptical surface features and the surrounding portion (bottom);

(25) FIG. **19B**, pertaining to Examples 15A-15B, is a graph illustrating that incorporating the secondary surface features over the entire textured region resulted in a lowered pixel power deviation compared to substrates where the secondary surface features were incorporated only on the elliptical surface features;

(26) FIG. **19C**, pertaining to Examples 15A-15B, is a graph illustrating that incorporating the secondary surface features over the entire textured region resulted in a higher transmission haze compared to substrates where the secondary surface features were incorporated only on the elliptical surface features;

(27) FIG. **19D**, pertaining to Examples 15A-15B, is a graph illustrating that incorporating the secondary surface features over the entire textured region did not substantially affect specular reflectance compared to substrates where the secondary surface features were incorporated only on the elliptical surface features;

(28) FIG. **19E**, pertaining to Examples 15A-15B, is a graph illustrating that incorporating the secondary surface features over the entire textured region slightly affected specular reflectance compared to substrates where the secondary surface features were incorporated only on the elliptical surface features, and increasingly so as wavelength deviated from about 455 nm;

(29) FIG. **20A**, pertaining to Example 16, are white light interferometer graphs illustrating the topography of the elliptical surface features and the surrounding portion (top) and the secondary surface features (bottom) disposed at the elliptical surface features (left) and the surrounding portion (right); and

(30) FIG. **20B**, pertaining to Example 16, are atomic force microscopy images of the secondary surface features disposed at an elliptical surface feature (left) and the surrounding portion (right), illustrating that the secondary surface features at the surrounding portion imparted a higher surface roughness ($R_{\text{sub.a}}$) than the at the elliptical surface feature (because the surrounding portion was not previously etched and thus more sensitive to the etching that imparted the secondary surface features).

DETAILED DESCRIPTION

(31) Referring now to FIG. **1**, a display article **10** includes a substrate **12**. In embodiments, the display article **10** further includes a housing **14** to which the substrate **12** is coupled and a display **16** within the housing **14**. In such embodiments, the substrate **12** at least partially covers the display **16** such that light that the display **16** emits can transmit through the substrate **12**.

(32) The substrate **12** includes a primary surface **18**, a textured region **20** defined on the primary surface **18**, and a thickness **22** that the primary surface **18** bounds in part. The primary surface **18** generally faces toward an external environment **24** surrounding the display article **10** and away from the display **16**. The display **16** emits visible light that transmits through the thickness **22** of the substrate **12**, out the primary surface **18**, and into the external environment **24**.

(33) Referring now to FIGS. **2-5**, in embodiments, the textured region **20** includes surface features **26**. The surface features **26** are randomly distributed. Each of the surface features **26** has a perimeter **28**. The perimeter **28** is parallel to a base-plane **30** extending through the thickness **22** of the substrate **12** below the textured region **20**. The perimeter **28** is elliptical. The base-plane **30** provides a conceptual reference and not a structural component.

(34) The textured region **20** further includes a surrounding portion **32**. In embodiments, the surface features **26** are set into the surrounding portion **32**. In those embodiments, the surface features **26** are elliptical blind holes that extend down into the thickness **22** of the substrate **12** toward the base-plane **30**. In other embodiments, the surface features **26** project from the surrounding portion **32**. In those embodiments, the surface features **26** are elliptical pillars that project from the surrounding portion **32** away from the base-plane **30** and toward the external environment **24**. The surrounding portion **32** forms a contiguous surface from which the surface features **26** either project from or are set into.

(35) As mentioned, the perimeter **28** of each of the surface features **26** is elliptical—that is, the shape of the perimeter **28** is an ellipsis. Each perimeter **28** thus has a longest dimension **34**, which is twice a longest axis from a center **36** of the ellipse. In addition, each perimeter **28** has a width **38**, which is twice a shortest axis from the center **36** of the ellipse. The incorporation of the surface features **26** having the perimeter **28** that is elliptical has a meaningful effect on anti-glare performance. Efforts leading to this disclosure also investigated surface features **26** having a circular perimeter **28**. Substrates incorporating circular surface features as the textured region tended to produce observable reflection color artifacts upon reflecting ambient light. Comparative Example 1A below further elaborates that point. The surface features **26** with the perimeter **28** that is elliptical, or that incorporate elliptical segments, reflect light with reduced color artifacts, as well as provide beneficial transmission haze, pixel power deviation, and distinctiveness-of-image values.

(36) As mentioned, the surface features **26** are randomly distributed. In other words, the surface features **26** are aligned with each other and the surface features **26** are not arranged in a pattern. For example, the longest dimensions **34** of the surface features **26** are not all parallel with each other (i.e., not all aligned to be extending in the same directions). In addition, no grouping of the surface features **26** is arranged geometrically the same as any other grouping of the surface features **26**.

(37) The textured region **20** further includes (i) one or more higher surfaces **40** that reside at a higher mean elevation **42** from the base-plane **30** and (ii) one or more lower surfaces **41** that reside at a lower mean elevation **44** from the base-plane **30**. The lower mean elevation **44** is closer to the base-plane **30** than the higher mean elevation **42**. The one or more higher surfaces **40** all reside at the same elevation within manufacturing tolerances. The one or more lower surfaces **41** all reside at the same elevation within manufacturing tolerances. In embodiments, the surface features **26** provide the one or more higher surfaces **40** and the surrounding portion **32** provides the one or more lower surfaces **41**. In such circumstances, the surface features **26** project from the surrounding portion **32**. In embodiments, the surface features **26** provide the one or more lower surfaces **41** and the surrounding portion **32** provides the one or more higher surfaces **42**. In such circumstances, the surface features **26** are set into the surrounding portion **32**.

(38) In embodiments, the longest dimension **34** of each surface feature **26** is 5 μm , 10 μm , 15 μm , 20 μm , 25 μm , 30 μm , 40 μm , 50 μm , 60 μm , 70 μm , 80 μm , 90 μm , 100 μm , 110 μm , 120 μm , 130 μm , 140 μm , 150 μm , or within any range bounded by any two of those values (e.g., 100 μm to 110 μm , 15 μm to 40 μm , 5 μm to 15 μm , and so on). In embodiments, the longest dimension **34** of each surface features **26**, on average, is about 100 μm .

(39) In embodiments, the surface features **26** have a fill-fraction of the textured region **20** of 30%, 40%, 50%, 60%, 70%, or within any range defined by any two of those values (e.g., 40% to 60%, and so on). The perimeter **28** of each surface feature **26** encompasses an area of a planar cross-section of the textured region **20** parallel to the base-plane **30**. The sum of the areas of all of the perimeters **28** of all of the surface features **26** divided by the total area of the planar cross-section of the textured region **20** is the “fill-fraction” of the surface features **26**. It is believed that a fill-fraction of about 50% suppresses specular reflection. In other words, some the ambient light reflects off the surface features **26**, and some of the light reflects off the surrounding portion **32**. A fill-fraction of about 50% provides maximum destructive interference between the ambient light

reflecting off the surface features **26** and the ambient light reflecting off the surrounding portion **32**. (40) In embodiments, a distance **46** separates the center **36** of one surface feature **26** from the center **36** of the next nearest surface features **26**. This document may refer to the distance **46** as “the pitch” of the surface features **26**. In embodiments, the average of these distances **46** throughout the textured region **20** is 80 μm , 90 μm , 100 μm , 110 μm , 120 μm , 130 μm , 140 μm , 150 μm , or within any range bounded by any two of those values (e.g., 90 μm to 120 μm , 100 μm to 130 μm , and so on). It is believed that to cause the textured region **20** to reflect light where the peak intensity of reflected light is at a small angle (such as about 0.3 degrees), then the distance **46** separating the centers **36** of the surface features **26** should be within the range of about 90 μm to 120 μm . For example, considering the 550 nm wavelength portion of reflected ambient light, when the distance **46** is 105 μm , then the scattering angle providing the peak intensity of reflected light (at 550 nm wavelength) ought to be 0.3 degrees (because $\text{pitch} \cong \text{wavelength} / \sin(\theta)$).

(41) In embodiments, the lower mean elevation **44** provided by the one or more lower surfaces **41** differs from the higher mean elevation **42** provided by the one or more higher surfaces **40** by a distance **48** of 0.05 μm , 0.10 μm , 0.15 μm , 0.20 μm , 0.25 μm , 0.30 μm , 0.35 μm , 0.40 μm , 0.45 μm , 0.50 μm , 0.55 μm , 0.60 μm , 0.65 μm , 0.70 μm or within any range defined by any two of those values (e.g., 0.10 μm to 0.20 μm , 0.05 μm to 0.70 μm , 0.05 μm to 0.60 μm , and so on). When the surface features **26** project from the surrounding portion **32**, the difference **48** is a height of the surface features **26** from the surrounding portion **32**. When the surface features **26** are set into the surrounding portion **32**, the difference **48** is a depth of the surface features **26** from the surrounding portion **32**.

(42) Referring now to FIGS. 6-8C, in embodiments of the textured region **20**, the surface features **26** include or are separated into larger surface features **26L** and smaller surface features **26S**. “Larger” and “smaller” here are relative terms meaning that the larger surface features **26L** are larger than the smaller surface features **26S**. Each larger surface features **26L** and each smaller surface feature **26S** have a perimeter **28L**, **28S**, respectively, which is elliptical. Each perimeter **28L**, **28S** has a longest dimension **34L**, **34S**, respectively. The longest dimensions **34L** of the larger surface features **26L** fall within a range of longest dimensions **34L**. The longest dimensions **34S** of the larger surface features **26S** fall within a range of longest dimensions **34S**, that are exclusive of the range of longest dimensions **34L** of the larger surface features **26L**. The longest dimensions **34L** of the range of longest dimensions **34L** of the larger surface features **26L** are longer than the longest dimensions **34S** of the range of longest dimensions **34S** of the smaller surface features **26S**.

(43) As mentioned above, in an attempt to force the intensity of reflected light off the textured region **20** to peak at about 0.3 degrees, the distance **46** between the centers **36** of the surface features **26** ought to be about 105 μm . As additionally mentioned above, to maximize specular reflection suppression via destruction interference, the fill-fraction of the surface features **26** ought to be about 50%. To accommodate both of those criteria, then the longest dimensions **34** of the surface features **26** ought to be 100 μm . However, surface features **26** having such a longest dimension **34** on the 100 μm scale have a tendency to generate relatively high pixel power deviation. The addition of the smaller surface features **26S** mitigates that consequence and generally results in a more acceptable, lower, pixel power deviation. The incorporation of the smaller surface features **26S** can improve the pixel power deviation compared to if only the larger surface features **26L** were incorporated.

(44) In embodiments, the perimeters **28L** of the larger surface features **26L** totally surround the perimeters **28S** of some of the smaller surface features **26S**. In embodiments, the perimeters **28S** of some or all of the smaller surface features **26S** reside entirely outside of the perimeters **28L** of the larger surface features **26L**. In embodiments, the perimeter **28S** of any given smaller surface features **26S** partially overlaps with the perimeter **28L** of any given larger surface feature **28L**, such that (i) part of the perimeter **28S** of the smaller surface features **26S** is inside the perimeter **28L** of the larger surface feature **26L** and (ii) part of the perimeter **28S** of the smaller surface features **26S**

is outside the perimeter **28L** of the larger surface feature **26L**. Further, although not separately illustrated, the smaller surface features **26S**, instead of having the perimeter **28S** that is elliptical, have the perimeter **28S** that can be circular.

(45) As mentioned, in embodiments, the textured region includes the two elevations (see, e.g., FIG. **8C**)—the higher mean elevation **42** and the lower mean elevation **44**. In embodiments, the larger surface features **26L** provide at least a portion of the one or more higher surfaces **40** residing at the higher mean elevation **42**. For example, in embodiments, the larger surface features **26L** project from the surrounding portion **32**, some of the smaller surface features **26S** project from the surrounding portion **32**, and the remaining smaller surface features **26S**, those that do not project from the surrounding portion **32**, are set into the larger surface features **26L**. In such embodiments, (i) the larger surface features **26L** and (ii) the smaller surface features **26S** that project from the surrounding portion **32** provide the one or more higher surfaces **40** residing at the higher mean elevation **42**. The surrounding portion **32** and the smaller surface features **26S** that are set into larger surface features **26L** provide the one or more lower surfaces **41** residing at the lower mean elevation **44**. Such a configuration can be the result of a one step-etching process. A single etching step etches away the substrate **12** to form all of (i) the larger surface features **26L** projecting from the surrounding portion **32**, (ii) the smaller surface features **26S** projecting from the surrounding portion **32**, and (iii) the smaller surface features **26S** set into the larger surface features **26L**.

(46) In embodiments (not separately illustrated, but the inverse of FIG. **8C**), the larger surface features **26L** provide at least a portion of the one or more lower surfaces **41** residing at the lower mean elevation **44**. For example, in embodiments, the larger surface features **26L** are set into the surrounding portion **32**, some of the smaller surface features **26S** are set into the surrounding portion **32**, and the remaining smaller surface features **26S**, those that are not set into the surrounding portion **32**, project from the larger surface features **26L**. In such embodiments, (i) the larger surface features **26L** and (ii) the smaller surface features **26S** that are set into the surrounding portion **32** provide the one or more lower surfaces **41** residing at the lower mean elevation **44**. The surrounding portion **32** and the smaller surface features **26S** that project from the larger surface features **26L** provide the one or more higher surfaces **40** residing at the higher mean elevation **42**. Such a configuration can be the result of a one step-etching process.

(47) In embodiments (see FIGS. **8A** and **8B**), the textured region **20** further includes one or more intermediate surfaces **43** (e.g., **43a**, **43b**) that reside at one or more intermediate mean elevations **45** (e.g., **45a**, **45b**) from the base-plane **30**. The one or more intermediate mean elevations **45** (e.g., **45a**, **45b**) are between the higher mean elevation **42** and the lower mean elevation **44**. These embodiments of the textured region **20** thus have surfaces at three or four distinct elevations from the base-plane **30**.

(48) For example (see FIG. **8A**), in embodiments, the larger surface features **26L** provide at least a portion of the one or more higher surfaces **40** residing at the higher mean elevation **42**. In these embodiments, the larger surface features **26L** project from the surrounding portion **32**. The surrounding portion **32** provides the one or more intermediate surfaces **43b** that reside at the intermediate elevation **45b**. In addition, some of the smaller surface features **26S** are set into the surrounding portion **32** and provide the one or more lower surfaces **41** that reside at the lower mean elevation **44**. The remaining smaller surface features **26S**, those that are not set into the surrounding portion **32**, are set into the larger surface features **26L**, and provide the one or more intermediate surfaces **43a** residing at the intermediate mean elevation **45a**. Such a configuration can be the result of a two-step etching process. A first etching step etches away the substrate **12** to form the surrounding portion **32** and the larger surface features **26L** projecting therefrom. A second etching step etches away the substrate **12** to form the smaller surface features **26S** set into the larger surface features **26L** and the surrounding portion **32**.

(49) Note that, when the depth of substrate **12** removed during the second etching step is equal to the depth of substrate **12** that was removed during the first etching step, the textured region **20**

would only include three distinct elevations—the higher mean elevation **42**, the lower mean elevation **44**, and a single intermediate mean elevation **45**. As applied to the example of FIG. **8A**, both the smaller surface features **26S** set into the larger surface features **26L** and the surrounding portion **32** would provide the one or more intermediate surfaces **43** residing at the single intermediate mean elevation **45**.

(50) As another example (see FIG. **8B**), the larger surface features **26L** provide at least a portion of the one or more lower surfaces **41** disposed at the lower mean elevation **44**. In these embodiments, the larger surface features **26L** are set into the surrounding portion **32**. The surrounding portion **32** provides the one or more intermediate surfaces **43b** disposed at the intermediate mean elevation **45b**. In addition, some of the smaller surface features **26S** project from the larger surface features **26L**, and provide the one or more intermediate surfaces **43a** disposed at the intermediate mean elevation **45a**. The remaining smaller surface features **26S**, those that do not project from the from the larger surface features **26L**, project from the surrounding portion **32**, and provide the one or more higher surfaces **40** residing at the higher mean elevation **42**. Again, the textured region **20** includes four distinct elevations **42**, **45a**, **45b**, and **44**. Such a configuration can be the result again of a two-step etching process. A first etching step etches away the substrate **12** to form the larger surface features **26L** set into the surrounding portion **32**. A second etching step etches away the substrate **12** to form the smaller surface features **26S** projecting from the larger surface features **26L** and the surrounding portion **32**. Note that, when the depth of substrate **12** removed during the second etching step is equal to the depth of substrate **12** that was removed during the first etching step, the textured region **20** would only include three distinct elevations—the higher mean elevation **42**, the lower mean elevation **44**, and a single intermediate mean elevation **45**.

(51) In embodiments, in additional reference to FIGS. **9A-10D**, the textured region **20** includes a first portion **58** and a second portion **60**. Either the first portion **58** or the second portion **60** provides the one or more higher surfaces **40** residing at the higher mean elevation **42**. The other of the first portion **58** and the second portion **60** provides the one or more lower surfaces **41** residing at the lower mean elevation **44**. The textured region **20** in these embodiments thus provide two distinct elevations **42**, **44**. The first portion **58** is everything colored white in the examples of FIGS. **9A-10B** and the second portion **60** is everything colored black. The opposite scenario is just as valid—the first portion **58** could be everything that is black and the second portion **60** could be everything that is white.

(52) As perhaps best illustrated in FIGS. **9B** and **10B**, elliptical perimeters **28** define the first portion **58**, except for overlapping portions **66** of the elliptical perimeters **28**. Two ellipses **28** in a plane are said to overlap if their area intersection is non-empty. The overlapping portions **66** of the elliptical perimeters **28** define, in part, the second portion **60**. For example, referring specifically to FIG. **9B**, elliptical perimeter **28A** does not overlap any other elliptical perimeter **28**. Thus, everything that the elliptical perimeter **28A** bounds is part of the first portion **58**. Moving to perimeter **28B**, perimeter **28B** is entirely overlapped (i.e., surrounded) by perimeter **28C**. Thus, everything that the elliptical perimeter **28B** bounds is part of the second portion **60**. Now considering perimeter **28D**, perimeter **28D** only partially overlaps with perimeter **28C**—part of perimeter **28D** is inside perimeter **28C** and part of perimeter **28D** is outside perimeter **28C**. The part of perimeter **28D** that is outside perimeter **28C** is not an overlapping portion **66** and thus forms part of the first portion **58**. However, the part of perimeter **28D** that is inside perimeter **28C** is an overlapping portion and thus forms part of the second portion **60**. Finally, looking at perimeter **28C**, perimeter **28C** fully overlaps perimeter **28B**, partially overlaps perimeter **28D**, and fully or partially overlaps several other perimeters. Those portions of perimeter **28C**, overlapping other perimeters **28**, form part of the second portion **60**. The remainder of perimeter **28C** does not overlap any other perimeter **28** and thus forms part of the first portion **58**.

(53) In addition to the overlapping portions **66** of the elliptical perimeters **28**, the second portion **60** of the textured region **20** further comprises any part of the textured region **20** that is not the first

portion **58**. More specifically, any parts **70** of the textured region **20** outside of any of the elliptical perimeters **28** form part of the second portion **60**.

(54) Like the embodiments previously discussed, each of the elliptical perimeters **28** of the textured region **20** is randomly distributed and randomly oriented. Each of the elliptical perimeters **28** comprises a longest dimension **34**. The longest dimensions **34** of the elliptical perimeters **28** are not all parallel to each other. Thus, the longest dimensions **34** of the elliptical perimeters **28** are not all commonly aligned. The random distribution and orientation of the elliptical perimeters **28** and lack of pattern prevent or reduce the ability of the textured region **20** to generate Moiré interference fringes and other optical distortions upon reflecting ambient light.

(55) As mentioned, referring to FIGS. **10C** and **10D**, either the first portion **58** or the second portion **60** provides the one or more higher surfaces **40** residing at the higher mean elevation **42**, while the other of the first portion **58** and the second portion **60** provides the one or more lower surfaces **41** residing at the lower mean elevation **44**. In embodiments, such as illustrated at FIG. **10C**, the first portion **58** provides the one or more higher surfaces **40** residing at the higher mean elevation **42**. The second portion **60** provides the one or more lower surfaces **41** residing at the lower mean elevation **44**. Stated another way, in these embodiments, the first portion **58** projects out toward the external environment **24** from the second portion **60**.

(56) In other embodiments, such as illustrated at FIG. **10D**, the first portion **58** provides the one or more lower surfaces **41** residing at the lower mean elevation **42**. The second portion **60** provides the one or more higher surfaces **40** residing at the higher mean elevation **44**. Stated another way, in these embodiments, the second portion **60** projects out toward the external environment **24** from the first portion **58**.

(57) The higher mean elevation **44** differs from the lower mean elevation **42**, in these embodiments with the first portion **58** and the second portion **60**, by the distance **48**, which again is 0.05 μm , 0.10 μm , 0.15 μm , 0.20 μm , 0.25 μm , 0.30 μm , 0.35 μm , 0.40 μm , 0.45 μm , 0.50 μm , 0.55 μm , 0.60 μm , 0.65 μm , 0.70 μm or within any range defined by any two of those values (e.g., 0.10 μm to 0.20 μm , 0.05 μm to 0.70 μm , 0.05 μm to 0.60 μm , and so on). A one-step etching process can produce embodiments including the first portion **58** and the second portion **60** as described.

(58) In embodiments, the elliptical perimeters **28** are divided by size into two different subsets—larger elliptical perimeters **28L** and smaller elliptical perimeters **28S**, which are smaller than the larger elliptical perimeters **28L**. The larger elliptical perimeters **28L** are larger than the smaller elliptical perimeters **28S** in the sense that the longest dimensions **34** of the larger elliptical perimeters **28L** are part of a range of longest dimensions **34** that is longer than a range of longest dimensions **34** of which the longest dimensions **34** of the smaller elliptical perimeters **28S** are a part. For example, referring to FIG. **9B**, the longest dimension **34** of perimeter **28C**, belonging to the larger elliptical perimeters **28L**, is longer than the longest dimensions **34** of perimeters **28A**, **28B**, and **28D**, which all belong to the smaller elliptical perimeters **28S**. The first portion **58** of the textured region **20** having the first elevation **62** from the base-plane **30** is bounded by (i) the smaller elliptical perimeters **28S** that do not overlap or intersect with the larger elliptical perimeters **28L**, (ii) portions of the smaller elliptical perimeters **28S** outside of the first elliptical perimeters **28S** that partially overlap with the larger elliptical perimeters **28L**, and (iii) portions of the larger elliptical perimeters **28L** that do not overlap with the smaller elliptical perimeters **28S**. The second portion **60** of the textured region **20** residing at the second elevation **64** is everything else that is not the first portion **58** of the textured region **20**.

(59) As discussed above, incorporation of relatively large surface features **28** that are spaced relatively far apart but with a fill-factor of about 50% can force intensity of reflected light to peak at about 0.3 degrees while maximizing interferometric suppression of specular reflection. However, the same result can be achieved if relatively small surface features **26** of two size ranges are utilized and the difference between the two size ranges is relatively small.

(60) In embodiments, as illustrated at FIGS. **9A** and **9B**, at least some of the larger elliptical

perimeters **28L** entirely encompass more than one of the smaller elliptical perimeters **28S**. Note that in some embodiments, such as that illustrated at FIG. **6**, at least some of the larger elliptical perimeters **28L** encompass more than one of the smaller elliptical perimeters **28S** but without any of the larger elliptical perimeters **28S** only partially overlapping any of the smaller elliptical perimeters **28S**. The elimination of smaller elliptical perimeters **28S** that would only partially overlap the larger elliptical perimeter **28L** is thought to reduce large angle scattering of reflected light. Large angle scattering can increase haze. In addition, the lithography masks that incorporate the designs that permit etching of such a textured region **20** have a resolution limit—too small a structure, such as where a small elliptical perimeter **28S** only partially overlaps a larger elliptical perimeter **28L**, can cause aliasing if sampled with insufficient resolution.

(61) In embodiments, a fill-fraction of the larger elliptical perimeters **28L** is within a range of 40% to 60%, while a fill-fraction of the smaller elliptical perimeters **28S** is within a range of 10% to 30%. The textured region **20** occupies an area **72**. The fill-fraction of the larger elliptical perimeters **28L** here is the percentage of the area **72** encompassed by all of the larger elliptical perimeters **28L**. The fill-fraction of the smaller elliptical perimeters **28S** here is the percentage of the area **72** encompassed by all of smaller elliptical perimeters **28S**. The smaller elliptical perimeters **28S** having a lower fill-fraction limits the impact of the smaller elliptical perimeters **28S** on the scattering spectrum of the overall design of the textured region **20**.

(62) In embodiments (see, e.g., FIG. **3**), the textured region **20** further includes one or more sections **80** that have secondary surface features **82**. The secondary surface features **82** impart a surface roughness to the one or more sections **80** of the textured region **20**. The increased surface roughness imparts surface scattering to the textured region **20**, which generally lowers pixel power deviation and distinctness of image. The surface roughness imparted is 5 nm, 10 nm, 15 nm, 20 nm, 25 nm, 30 nm, 35 nm, 40 nm, 45 nm, 50 nm, 55 nm, 60 nm, 65 nm, 70 nm, 80 nm, 90 nm, or 100 nm or within any range bounded by any two of those values (e.g., 5 nm to 100 nm, and so on). As used herein, surface roughness (R.sub.a) is measured with an atomic force microscope, such as an atomic force microscope controlled by a NanoNavi control station distributed by Seiko Instruments Inc. (Chiba, Japan), with a scan size of 5 μ m by 5 μ m. Surface roughness (R.sub.a), as opposed to other types of surface roughness values such as R.sub.q, is the arithmetical mean of the absolute values of the deviations from a mean line of the measured roughness profile.

(63) In embodiments, the one or more sections **80** that include the secondary surface features **50** include the one or more higher surfaces **40**, the one or more lower surfaces **41**, and if present, the one or more intermediate surfaces **43a**, **43b**. In embodiments, the secondary surface features **82** are disposed on the surface features **26** but not the surrounding portion **32**. In embodiments, the secondary surface features **82** are disposed on the surrounding portion **32** but not the surface features **26**. In embodiments, the secondary surface features **82** are disposed on both the surrounding portion **32** and the surface features **26**. In embodiments, the secondary surface features **82** are disposed on the first portion **58** and not the second portion **60**, or vice-versa. In embodiments, the secondary surface features **82** are disposed on the first portion **58** and the second portion **60**. In embodiments, the one or more sections **80** that includes the secondary surface features **82** is coextensive with the textured region **20** meaning that the secondary surface features **82** are disposed throughout the entirety of the textured region **20**. In embodiments, the surface roughness (R.sub.a) imparted by the second surface features **82** at the surface features **26** is less than the surface roughness at the surrounding portion **32**.

(64) In embodiments, the substrate **12** includes a glass substrate or a glass-ceramic substrate. In embodiments, the substrate **12** is a multi-component glass composition having about 40 mol % to 80 mol % silica and a balance of one or more other constituents, e.g., alumina, calcium oxide, sodium oxide, boron oxide, etc. In some implementations, the bulk composition of the substrate **12** is selected from the group consisting of aluminosilicate glass, a borosilicate glass, and a phosphosilicate glass. In other implementations, the bulk composition of the substrate **12** is

selected from the group consisting of aluminosilicate glass, a borosilicate glass, a phosphosilicate glass, a soda lime glass, an alkali aluminosilicate glass, and an alkali aluminoborosilicate glass. In further implementations, the substrate **12** is a glass-based substrate, including, but not limited to, glass-ceramic materials that comprise a glass component at about 90% or greater by weight and a ceramic component. In other implementations of the display article **10**, the substrate **12** can be a polymer material, with durability and mechanical properties suitable for the development and retention of the textured region **20**.

(65) In embodiments, the substrate **12** has a bulk composition that comprises an alkali aluminosilicate glass that comprises alumina, at least one alkali metal and, in some embodiments, greater than 50 mol % SiO₂, in other embodiments, at least 58 mol % SiO₂, and in still other embodiments, at least 60 mol % SiO₂, wherein the ratio (Al₂O₃ (mol %) B₂O₃ (mol %))/Σ alkali metal modifiers (mol %)>1, where the modifiers are alkali metal oxides. This glass, in particular embodiments, comprises, consists essentially of, or consists of: about 58 mol % to about 72 mol % SiO₂; about 9 mol % to about 17 mol % Al₂O₃; about 2 mol % to about 12 mol % B₂O₃; about 8 mol % to about 16 mol % Na₂O; and 0 mol % to about 4 mol % K₂O, wherein the ratio (Al₂O₃ (mol %)+B₂O₃ (mol %))/Σ alkali metal modifiers (mol %)>1, where the modifiers are alkali metal oxides.

(66) In embodiments, the substrate **12** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 61 mol % to about 75 mol % SiO₂; about 7 mol % to about 15 mol % Al₂O₃; 0 mol % to about 12 mol % B₂O₃; about 9 mol % to about 21 mol % Na₂O; 0 mol % to about 4 mol % K₂O; 0 mol % to about 7 mol % MgO; and 0 mol % to about 3 mol % CaO.

(67) In embodiments, the substrate **12** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 60 mol % to about 70 mol % SiO₂; about 6 mol % to about 14 mol % Al₂O₃; 0 mol % to about 15 mol % B₂O₃; 0 mol % to about 15 mol % Li₂O; 0 mol % to about 20 mol % Na₂O; 0 mol % to about 10 mol % K₂O; 0 mol % to about 8 mol % MgO; 0 mol % to about 10 mol % CaO; 0 mol % to about 5 mol % ZrO₂; 0 mol % to about 1 mol % SnO₂; 0 mol % to about 1 mol % CeO₂; less than about 50 ppm As₂O₃; and less than about 50 ppm Sb₂O₃; wherein 12 mol %≤Li₂O+Na₂O+K₂O≤20 mol % and 0 mol %≤MgO+Ca≤10 mol %.

(68) In embodiments, the substrate **12** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 64 mol % to about 68 mol % SiO₂; about 12 mol % to about 16 mol % Na₂O; about 8 mol % to about 12 mol % Al₂O₃; 0 mol % to about 3 mol % B₂O₃; about 2 mol % to about 5 mol % K₂O; about 4 mol % to about 6 mol % MgO; and 0 mol % to about 5 mol % CaO, wherein: 66 mol %≤SiO₂+B₂O₃+CaO≤69 mol %; Na₂O+K₂O+B₂O₃+MgO+CaO+SrO>10 mol %; 5 mol %≤MgO+CaO+SrO≤8 mol %; (Na₂O+B₂O₃)—Al₂O₃≤2 mol %; 2 mol %≤Na₂O+3Al₂O₃≤6 mol %; and 4 mol %≤(Na₂O+K₂O)—Al₂O₃≤10 mol %.

(69) In embodiments, the substrate **12** has a bulk composition that comprises SiO₂, Al₂O₃, P₂O₅, and at least one alkali metal oxide (R₂O), wherein $0.75 > [(P_2O_5 \text{ (mol \%)} + R_2O \text{ (mol \%)}) / M_2O_3 \text{ (mol \%)}] \leq 1.2$, where $M_2O_3 = Al_2O_3 + B_2O_3$. In embodiments, $[(P_2O_5 \text{ (mol \%)} + R_2O \text{ (mol \%)}) / M_2O_3 \text{ (mol \%)}] = 1$ and, in embodiments, the glass does not include B₂O₃ and $M_2O_3 = Al_2O_3$. The substrate **12** comprises, in embodiments: about 40 to about 70 mol % SiO₂; 0 to about 28 mol % B₂O₃; about 0 to about 28 mol % Al₂O₃; about 1 to about 14 mol % P₂O₅; and about 12 to

about 16 mol % R.sub.2O. In some embodiments, the glass substrate comprises: about 40 to about 64 mol % SiO.sub.2; 0 to about 8 mol % B.sub.2O.sub.3; about 16 to about 28 mol % Al.sub.2O.sub.3; about 2 to about 12 mol % P.sub.2O.sub.5; and about 12 to about 16 mol % R.sub.2O. The substrate **12** may further comprise at least one alkaline earth metal oxide such as, but not limited to, MgO or CaO.

(70) In some embodiments, the substrate **12** has a bulk composition that is substantially free of lithium; i.e., the glass comprises less than 1 mol % Li.sub.2O and, in other embodiments, less than 0.1 mol % Li.sub.2O and, in other embodiments, 0.01 mol % Li.sub.2O, and in still other embodiments, 0 mol % Li.sub.2O. In some embodiments, such glasses are free of at least one of arsenic, antimony, and barium; i.e., the glass comprises less than 1 mol % and, in other embodiments, less than 0.1 mol %, and in still other embodiments, 0 mol % of As.sub.2O.sub.3, Sb.sub.2O.sub.3, and/or BaO.

(71) In embodiments, the substrate **12** has a bulk composition that comprises, consists essentially of or consists of a glass composition, such as Corning® Eagle XG® glass, Corning® Gorilla® glass, Corning® Gorilla® Glass 2, Corning® Gorilla® Glass 3, Corning® Gorilla® Glass 4, or Corning® Gorilla® Glass 5.

(72) In embodiments, the substrate **12** has an ion-exchangeable glass composition that is strengthened by either chemical or thermal means that are known in the art. In embodiments, the substrate **12** is chemically strengthened by ion exchange. In that process, metal ions at or near the primary surface **18** of the substrate **12** are exchanged for larger metal ions having the same valence as the metal ions in the substrate **12**. The exchange is generally carried out by contacting the substrate **12** with an ion exchange medium, such as, for example, a molten salt bath that contains the larger metal ions. The metal ions are typically monovalent metal ions, such as, for example, alkali metal ions. In one non-limiting example, chemical strengthening of a substrate **12** that contains sodium ions by ion exchange is accomplished by immersing the substrate **12** in an ion exchange bath comprising a molten potassium salt, such as potassium nitrate (KNO.sub.3) or the like. In one particular embodiment, the ions in the surface layer of the substrate **12** contiguous with the primary surface **18** and the larger ions are monovalent alkali metal cations, such as Li.sup.+ (when present in the glass), Na.sup.+, K.sup.+, Rb.sup.+, and Cs.sup.+. Alternatively, monovalent cations in the surface layer of the substrate **12** may be replaced with monovalent cations other than alkali metal cations, such as Ag.sup.+ or the like.

(73) In such embodiments, the replacement of small metal ions by larger metal ions in the ion exchange process creates a compressive stress region in the substrate **12** that extends from the primary surface **18** to a depth (referred to as the “depth of layer”) that is under compressive stress. This compressive stress of the substrate **12** is balanced by a tensile stress (also referred to as “central tension”) within the interior of the substrate **12**. In some embodiments, the primary surface **18** of the substrate **12** described herein, when strengthened by ion exchange, has a compressive stress of at least 350 MPa, and the region under compressive stress extends to a depth, i.e., depth of layer, of at least 15 μm below the primary surface **18** into the thickness **22**.

(74) Ion exchange processes are typically carried out by immersing the substrate **12** in a molten salt bath containing the larger ions to be exchanged with the smaller ions in the glass. It will be appreciated by those skilled in the art that parameters for the ion exchange process, including, but not limited to, bath composition and temperature, immersion time, the number of immersions of the glass in a salt bath (or baths), use of multiple salt baths, additional steps such as annealing, washing, and the like, are generally determined by the composition of the glass and the desired depth of layer and compressive stress of the glass as a result of the strengthening operation. By way of example, ion exchange of alkali metal-containing glasses may be achieved by immersion in at least one molten bath containing a salt, such as, but not limited to, nitrates, sulfates, and chlorides, of the larger alkali metal ion. The temperature of the molten salt bath typically is in a range from about 380° C. up to about 450° C., while immersion times range from about 15 minutes up to about

16 hours. However, temperatures and immersion times different from those described above may also be used. Such ion exchange treatments, when employed with a substrate **12** having an alkali aluminosilicate glass composition, result in a compressive stress region having a depth (depth of layer) ranging from about 10 μm up to at least 50 μm , with a compressive stress ranging from about 200 MPa up to about 800 MPa, and a central tension of less than about 100 MPa.

(75) As the etching processes that can be employed to create the textured region **20** of the substrate **12** can remove alkali metal ions from the substrate **12** that would otherwise be replaced by a larger alkali metal ion during an ion exchange process, a preference exists for developing the compressive stress region in the display article **10** after the formation and development of the textured region **20**.

(76) In embodiments, the display article **10** exhibits a pixel power deviation (“PPD”). The details of a measurement system and image processing calculation used to obtain PPD values described in U.S. Pat. No. 9,411,180 entitled “Apparatus and Method for Determining Sparkle,” and the salient portions of which are related to PPD measurements are incorporated by reference herein in their entirety. Further, unless otherwise noted, the SMS-1000 system (Display-Messtechnik & Systeme GmbH & Co. KG) is employed to generate and evaluate the PPD measurements of this disclosure. The PPD measurement system includes: a pixelated source comprising a plurality of pixels (e.g., a Lenovo Z50 140 ppi laptop), wherein each of the plurality of pixels has referenced indices i and j ; and an imaging system optically disposed along an optical path originating from the pixelated source. The imaging system comprises: an imaging device disposed along the optical path and having a pixelated sensitive area comprising a second plurality of pixels, wherein each of the second plurality of pixels is referenced with indices m and n ; and a diaphragm disposed on the optical path between the pixelated source and the imaging device, wherein the diaphragm has an adjustable collection angle for an image originating in the pixelated source. The image processing calculation includes: acquiring a pixelated image of the transparent sample, the pixelated image comprising a plurality of pixels; determining boundaries between adjacent pixels in the pixelated image; integrating within the boundaries to obtain an integrated energy for each source pixel in the pixelated image; and calculating a standard deviation of the integrated energy for each source pixel, wherein the standard deviation is the power per pixel dispersion. As used herein, all PPD values, attributes and limits are calculated and evaluated with a test set-up employing a display device having a pixel density of 140 pixels per inch (PPI). In embodiments, the display article **10** exhibits a PPD of 0.8%, 0.9%, 1.0%, 1.1%, 1.2%, 1.3%, 1.4%, 1.5%, 1.6%, 1.7%, 1.8%, 1.9%, 2.0%, 2.25%, 2.5%, 2.75%, 3.0%, 3.25%, 3.5%, 3.75%, 4.0%, 4.25%, 4.5%, 4.75%, 5.0%, 5.25%, 5.5%, 5.75%, 6.0%, or within any range bounded by any two of those values (e.g., 0.8% to 2.0%, 0.9% to 2.25%, 2.0% to 6.0%, and so on). In embodiments, the display article **10** exhibits a PPD of less than 2.0%.

(77) In embodiments, the substrate **12** exhibits a distinctness-of-image (“DOI”). As used herein, “DOI” is equal to $100 \times (R_{\text{sub.S}} - R_{\text{sub.0.3}}) / R_{\text{sub.S}}$, where $R_{\text{sub.S}}$ is the first surface specular reflectance flux measured from incident light (at 20° from normal) directed onto the textured region **20**, and $R_{\text{sub.0.3}}$ is the reflectance flux measured from the same incident light at 0.3° from the specular reflectance flux, $R_{\text{sub.S}}$. The sample is optically coupled to flat black glass with index-matching oil on the back side to eliminate second-surface reflections. Unless otherwise noted, the DOI values and measurements reported in this disclosure are obtained according to the ASTM D5767-18, entitled “Standard Test Method for Instrumental Measurement of Distinctness-of-Image (DOI) Gloss of Coated Surfaces using a Rhopoint IQ Gloss Haze & DOI Meter” (Rhopoint Instruments Ltd.). In embodiments, the substrate **12** exhibits a distinctness-of-image (“DOI”) of 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, or within any range bounded by any two of those values (e.g., 20% to 40%, 10% to 96%, 35% to 60%, and so on).

(78) In embodiments, the substrate **12** exhibits a transmission haze. As used herein, the term “transmission haze” refers to the percentage of transmitted light scattered outside an angular cone

of about $\pm 2.5^\circ$ in accordance with ASTM D1003, entitled “Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics,” the contents of which are incorporated by reference herein in their entirety. Note that although the title of ASTM D1003 refers to plastics, the standard has been applied to substrates comprising a glass material as well. For an optically smooth surface, transmission haze is generally close to zero. In embodiments, the substrate **12** exhibits a transmission haze of 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1.0%, 1.5%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 12.5%, 15%, or within any range bounded by any two of those values (e.g., 0.3% to 15%, 0.9% to 1.0%, and so on).

(79) As used herein, “corrected color shift” is a measure of the amount of reflection color artifacts that the substrate **12** generates while reflecting ambient light from the textured region **20**. Referring now to FIG. **11**, to determine the corrected color shift, the substrate **12** with the textured region **20** to be tested is placed over the display **16**, with oil **74** disposed between the substrate **12** and the display **16** to suppress the light reflections of the surface of the substrate **12** facing the display **16** and the surface of the display **16**. The oil **74** has a refractive index matching a refractive index of the substrate **12**. Room lights **76** emit light as they normally would. A white light source **78** illuminates the substrate **12**. The textured region **20** of the substrate **12** faces toward the white light source **78**. Since the reflection color artifacts of the substrate **12** are much more easily observed and are more accurately measured when the display **16** is turned off, the display **16** is switched-off when color separation measurements are conducted. The textured region **20** reflects a portion of the light that the white light source **78** emits as a scattered light pattern. A color CCD camera **80** captures an image of the scattered light pattern. The image is then digitally processed, and chromaticity coefficients (C.sub.x and C.sub.y) along a selected straight line through the locations with maximum C.sub.x (or C.sub.y) and minimum C.sub.x (or C.sub.y) are calculated. Here, chromaticity coefficients C.sub.x and C.sub.y are defined as

$C_{\text{sub.x}} = P_{\text{sub.R}} / (P_{\text{sub.R}} + P_{\text{sub.G}} + P_{\text{sub.B}})$ and $C_{\text{sub.y}} = P_{\text{sub.G}} / (P_{\text{sub.R}} + P_{\text{sub.G}} + P_{\text{sub.B}})$ respectively, in which P.sub.R, P.sub.G, and P.sub.B are the powers (or intensities) of red, green, and blue light, respectively, at a location of the scattered light pattern detected by the color CCD camera **80**. Chromaticity is an objective specification of the quality of a color regardless of its luminance. The color shifts along the selected line, $\Delta C_{\text{sub.x}}$ and $\Delta C_{\text{sub.y}}$, are calculated as the difference between the maximum C.sub.x and the minimum C.sub.x for $\Delta C_{\text{sub.x}}$, and the difference between the maximum C.sub.y and the minimum C.sub.y for $\Delta C_{\text{sub.y}}$. The color shifts $\Delta C_{\text{sub.x}}$ and $\Delta C_{\text{sub.y}}$ are then corrected to account for the fact that the visibility of color change that human eyes see is relative to not only the color shifts ($\Delta C_{\text{sub.x}}$ and $\Delta C_{\text{sub.y}}$) but also an angle separation **82** between the locations of the maximum and minimum C.sub.x (for $\Delta C_{\text{sub.x}}$) and the maximum and minimum C.sub.y (for $\Delta C_{\text{sub.y}}$). These corrected color shifts are defined as

$$(80) \quad C_{x_corrected} = \frac{d}{d_{\theta_sub.r}} \frac{r}{x} \quad C_x \quad C_{y_corrected} = \frac{d}{d_{\theta_sub.r}} \frac{r}{y} \quad C_y$$

(81) The $d_{\theta_sub.r}$ is reference angle separation arbitrarily set at $d_{\theta_sub.r} = 0.84$ degrees. This reference angle is chosen from the angle between two adjacent measurement points of 455 point color and luminance measurement of a 300×110 mm display viewed at 500 mm distance. The $d_{\theta_sub.x}$ and $d_{\theta_sub.y}$ are the angle separations in degree between the locations of maximum and minimum for C.sub.x and C.sub.y respectively. When the corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ are each less than 0.3, it is assumed that human eyes cannot perceive any reflection color artifacts that the substrate **12** is producing. In embodiments, the substrate **12** exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ of 0.00, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, or within any range bounded by any two of those values (0.01 to 0.3, 0.05 to 1.0, and so on). In embodiments, the substrate **12** exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ of less than 0.3, less than 0.2, less than 0.1, or even 0.0.

(82) In embodiments, the substrate **12** exhibits a specular reflectance of 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40, 45%, 50%, 55%, 60%, or 65%, or within any range bounded by any two of those values (e.g., 5% to 65%, 5% to 30%, and so on). In embodiments, the substrate **12** exhibits a specular reflectance that is less than 30%, less than 25%, less than 20%, or less than 10%. Specular reflectance here, noted as “c-Rspec” in the Examples that follow, refers to the value obtained using a Rhopoint IQ goniophotometer, and are reported in gloss units (GU). 100 gloss units is defined to be the specular reflection intensity from a highly polished piece of flat black glass with refractive index 1.567. The values are indicative of how much specular reflection is measured when the sample is optically coupled with oil to a black glass of matching index to remove second-surface reflections.

(83) Referring now to FIG. **12**, a method **100** of forming the textured region **20** of the substrate **12** of the display article **10** is herein described. At a step **102**, the method **100** includes generating a random distribution of ellipses **104** within an area **106** of a plane **108**.

(84) At a step **110**, the method **100** further includes preparing a lithography mask **112**. The design of the lithography mask **112** matches the random distribution of ellipses **104** generated at the step **102**. The lithography mask **112** has an area **114**. The area **114** matches the area **112** of the plane **108** within which the ellipses **104** were randomly distributed. The lithography mask **112** includes material **116** throughout the area **114**. The lithography mask **112** further includes voids **118** through the material **116**. The random distribution of the ellipses **104** generated at the step **102** define either (a) the material **116** throughout the area **114** or (b) the voids **118** through the material **116**. In the illustration at FIG. **12**, the black portions outside of the ellipses **104** represents the material **116** of the lithography mask **112**, while the white portions inside of the ellipses **104** are the voids **118** through the material **116** of the lithography mask **112**. The lithography mask **112** can be prepared to have the opposite relationship—the inside of the ellipses **104** (e.g., the white in FIG. **12**) can represent the material **116** of the lithography mask **112**, while outside of the ellipses **104** (e.g., the black in FIG. **12**) can represent the voids **118** through the material **116** of the lithography mask **112**.

(85) A workpiece **120** is prepared that incorporates the lithography mask **112**. The workpiece **120** includes the substrate **12**, lithography ink **122** on the primary surface **18** of the substrate **12** at which the textured region **20** is desired to be formed, and the lithography mask **112** over the lithography ink **122**.

(86) At a step **124**, the method **100** further includes exposing the workpiece **120** to a curing agent **126**. The curing agent **126** can cure the lithography ink **122**. The curing agent **126** can be ultraviolet light, among other things. The lithography mask **112** allows for selective curing of the lithography ink **122**. The curing agent **126** transmits through the voids **118** in the material **116** of the lithography mask **112** to cure exposed portions of the lithography ink **122**. However, the material **116** of the lithography mask **112** blocks non-exposed portions of the lithography ink **122** from exposure to the curing agent **126**. Thus, the non-exposed portions of the lithography ink **122** are not cured. After curing, the lithography mask **112** and the non-exposed portions of the lithography ink **122** are removed from the substrate **12**. The exposed portions of the lithography ink **122** remain on the substrate **12** in cured formed as an etching mask **128**.

(87) The random distribution of ellipses **104** generated at the step **102** defined the material **116** and the voids **118** through the material of the lithography mask **112**, and thus, the etching mask **128** on the substrate **12**. In other words, the etching mask **128** is either shaped to match the inside of the ellipses **104**, or shaped to match the outside of the ellipses **104** generated at the step **102**. The black inside the ellipses **104** of the illustration at FIG. **12** pertaining to a step **130** represents the etching mask **128**. The white outside of the ellipses **104** is the primary surface **18** of the substrate **12**. The opposite scenario would have existed if the lithography mask **112** was formed in the opposite manner.

(88) At the step **130**, the method **100** further includes contacting the substrate **12** with the etching mask **128** with an etchant **132**. The etchant **132** selectively etches the etching mask **128** because the

etching mask **128** allows the etchant **132** to contact some portions of the primary surface **18** of the substrate **12** and prevents the etchant **132** from contacting other portions of the primary surface **18** of the substrate **12**. Depending on how the lithography mask **112** was formed, the etching mask **128** either (i) allows the etchant **132** to etch into the substrate **12** inside the ellipses **104** in accordance with the random generation thereof from the step **102** and prevents the etchant **132** from etching into the substrate **12** outside of the ellipses **104**, or (ii) prevents the etchant **132** from etching into the substrate **12** inside the ellipses **104** in accordance with the random generation thereof from the step **102** and allows the etchant **132** to etch into the substrate **12** outside of the ellipses **104**.

Thereafter, the etching mask **128** is removed. The substrate **12** now has the textured region **20** on the primary surface **18** thereof, with the topography of the textured region **20** matching the random distribution of ellipses **104** generated at the step **102**.

(89) In embodiments, the method **100** further comprises repeating the steps **102**, **110**, **124**, **130** so that the substrate **12** with the textured region **20** is etched again. To elaborate, a random distribution of second ellipses **104** (different than the random distribution of the ellipses **104** from the initial step **102**) is generated within an area **106** of a second plane **108**. These second ellipses **104** can have a range of longest dimensions **34** that are different (e.g., smaller) than a range of longest dimensions **34** of the ellipses **104** previously generated. A second lithography mask **112** is then prepared that incorporates the second ellipses **104** either as the material **116** or the voids **118** through the material **116**. A second workpiece **120** is prepared that includes the substrate **12**, new lithography ink **122** disposed on the textured region **20** of the substrate **12**, and the second lithography mask **112** disposed on the new lithography ink **122**. The second workpiece **120** is then exposed to a curing agent **126**. The curing agent **126** transmits through the voids **118** through the material **116** of the second lithography mask **112** to cure exposed portions of the new lithography ink **122**. The material **116** of the second lithography mask **112** blocks non-exposed portions of the new lithography ink **122** from exposure to the curing agent **126**. Thus, the non-exposed portions of the new lithography ink **122** are not cured and are removed along with the second lithography mask **112**. The exposed portions of the new lithography ink **122** remain on the substrate **12** as a second etching mask **128**. The substrate **12** with the second etching mask **128** is then contacted with an etchant **132**. Such two-step etching processes can form the textured region **20** with three or four elevations **50-56** from the base-plane **30** as explained in connection with FIGS. **8A-8C**.

(90) In embodiments, referring to FIG. **13**, the step **102** of generating the random distribution of the ellipses **104** includes first distributing points **134** randomly within the area **106** of the plane **108**. Each of the points **134** so randomly distributed is separated by a minimum distance **136**. The points **134** can be randomly distributed pursuant to a random distribution algorithm, such as a Poisson disk distribution algorithm. Other algorithms such as maxi-min are possible as well. However, some distribution algorithms such as maxi-min tend to arrange the points **134** too hexagonally, which causes the resulting ellipses **104** to be very similar to one another in terms of major and minor axis lengths and eccentricities. This can lead to color artifacts. The Poisson disk distribution algorithm enforces the minimum distance **136** between the distributed points **134** but does not produce highly hexagonal arrangements.

(91) More specifically, Poisson disk sampling inserts a first point **134** into the area **106**. Then the algorithm inserts a second point **134** within the area **106**, placing the center at a random point within the area **106**. If the placement of the second point **134** satisfies the minimum distance **136** from the first point **134**, then the second point **134** stays in the area **106**. The algorithm then repeats this process until no more such points **134** can be placed within the area **106** that satisfies the minimum distance **136**. The result is a random distribution, but specific placement, of the points **134**.

(92) The maxi-min spacing algorithm is so named because it attempts to maximize the minimum nearest-neighbor distance **136** of a point **134** distribution. Because it proceeds iteratively, moving each point **134** to another place where it is further from any neighbors, the algorithm usually does

not achieve a perfect hexagonal lattice. It produces a random distribution with a relatively high degree of mean hexagonality, often exceeding 90%.

(93) After the points **134** are randomly distributed within the area **106** of the plane **108**, the points **134** are triangulated. By triangulating the points **134**, each point **134** is made a vertex of a triangle **138**. A plurality of triangles **138** are thus formed, none of which overlap. For example, the points **134** can be triangulated via a Delaunay triangulation. Other triangulation methods are possible. Some of those would likely produce a larger range of longest dimensions **34** for the surface features **26**.

(94) After the points **134** have been triangulated, an ellipse **104** is drawn inside each triangle **138** of the plurality of triangles **138**, resulting in a plurality of ellipses **104** in the area **106** of the plane **108**. For example, each ellipse **104** can be a Steiner inellipse. An inellipse is an ellipse **104** that touches the three sides of the triangle **138** in which the ellipse **104** is drawn. A Steiner inellipse is an ellipse **104** that touches the triangle **138** at the midpoint of the sides of the triangle **138** and provides the maximum area for the ellipse **104** (a Steiner inellipse occupies $\pi\sqrt{3}/9$ or about 60.46% of the area of the triangle **138**, regardless of the triangle dimensions). The center of Steiner inellipse is the centroid (i.e., the point where the three medians of the triangle coincide) of the triangle **138** within which it is drawn.

(95) After the plurality of ellipses **104** have been drawn, the points **134** and the triangles **138** are removed from the area **106** of the plane **108**. Only the ellipses **104** that were drawn in the triangles **138** remain on the area **106** of the plane **108**. In embodiments, the sizes of the ellipses **104** are scaled down so that the fill-fraction of the ellipses **104** in the area **106** of the plane **108** is within a range of 45% to 55%. The fill-fraction here is the percentage of the area **106** of the plane **108** that the ellipses **104** occupy. As mentioned, if Steiner inellipses are drawn, then the fill-factor of the ellipses **104** before scaling down is about 60.46%. In embodiments, the sizes of the ellipses **104** are scaled down so that the fill-fraction of the ellipses **104** in the area **106** of the plane **108** is 50%. Each ellipse **104** has a longest dimension **140**, and to scale down the ellipses **104**, the longest dimension **140** of each of the ellipses **104** can be decreased by a percentage that is identical, with the centers of the ellipses **104** remaining in the same position.

(96) Referring now to FIG. **14**, another method **200** of forming the textured region **20** of the substrate **12** of the display article **10** is herein disclosed. At a step **202**, the method **200** includes generating a random distribution of first ellipses **204** within a first area **206**. Similarly, at a step **208**, the method **200** further includes generating a random distribution of second ellipses **210** within a second area **212**. Longest dimensions **214** of first ellipses **204** are longer than longest dimensions **216** of the second ellipses **210**.

(97) The steps **202**, **208** thus are generally the same as the step **102** of the method **100** and can be executed in the same manner. In other words, and referring back to FIG. **13**, the steps **202**, **208** of generating the random distribution of the first ellipses **204** and generating the random distribution of the second ellipses **210** both separately include (i) distributing points **134** randomly within an area **106** of a **108**, each of the points **134** separated by a minimum distance **136**; (ii) triangulating the points **134** so that each point is made a vertices of a triangle **138** thus forming a plurality of triangles **138**, and none of the triangles **138** overlap; (iii) drawing an ellipse **104** inside each triangle **138** of the plurality of triangles **138**; and removing the points **134** and the triangles **138** so that only the ellipses **104** that were drawn in the triangles **138** remain on the area **106** of the plane **108**.

(98) With the method **200**, there is the further caveat that the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the first ellipses **204** is greater than the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the second ellipses **210**. The longer minimum distance **136** for the points **134** from which the first ellipses **204** are generated results in the longest dimensions **140** of the first ellipses **204** being longer than the longest dimensions **140** of the second ellipses **210**. In

embodiments, the minimum distance separating each of the points **134** of the plane **108** for the random distribution of the first ellipses **204** is 15 μm , 16 μm , 17 μm , 18 μm , 19 μm , 20 μm , 30 μm , 40 μm , 50 μm , 60 μm , 70 μm , 80 μm , 90 μm , 100 μm , 110 μm , 120 μm , 130 μm , 140 μm , or within a range bounded by any two of those values (e.g., 100 μm to 120 μm , 80 μm to 130 μm , and so on). In embodiments, the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the second ellipses **210** is within 13 μm , 14 μm , 15 μm , 16 μm , 17 μm , 18 μm , 19 μm , 20 μm , 25 μm , 30 μm , 35 μm , 40 μm , or within any range bounded by any two of those values (e.g., 20 μm to 30 μm , 15 μm to 40 μm and so on).

(99) In embodiments, the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the first ellipses **204** is less than or equal 20 μm . In these embodiments, the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the second ellipses **210** is less than or equal 18 μm . The minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the first ellipses **204** differs from the minimum distance **136** separating each of the points **134** of the plane **108** for the random distribution of the second ellipses **210** by a value within a range of 1 μm to 3 μm .

(100) At a step **218**, the method **200** further includes forming a new area **220** that superimposes the first ellipses **204** of the first area **206** and the second ellipses **210** of the second area **212**. The new area **220** is the combination of the first ellipses **204** and the second ellipses **210** within the same new area **220**. The new area **220** includes free portions **222** of the first ellipses **204** and the second ellipses **210** that do not overlap. The free portions **222** would include any of the first ellipses **204** and the second ellipses **210** that do not overlap at all, and the portions of the first ellipses **204** and the second ellipses **210** that are not overlapped. The new area **220** further includes overlapping portions **224** of the first ellipses **204** and the second ellipses **210**. The overlapping portions **224** are where the first ellipses **204** and the second ellipses **210** occupy the same space of the new area **220**. Finally, the new area **220** further includes empty portions **226** where neither the first ellipses **204** nor the second ellipses **210** are present. The empty portions **226** are outside of the first ellipses **204** and outside of the second ellipses **210**.

(101) In embodiments, the method **200** further includes removing from the new area **220** any of the second ellipses **210** that the first ellipses **204** partially but not fully overlap. Such a new area **220** would appear like FIG. 6, where all of the smaller ellipses are either fully inside the larger ellipses or fully outside of the larger ellipses. None of the smaller ellipses are partially inside and partially outside of the larger ellipses. Those smaller ellipses are removed in these embodiments.

(102) At a step **228**, the method **200** further includes preparing a lithography mask **112**. The lithography mask **112** includes an area **114**, material **116** within the area **114**, and voids **118** through the material **116**. The material **116** within the area **114** is defined by (i.e., matches) the either (i) the free portions **222** of the first ellipses **204** and the second ellipses **210** that do not overlap or (ii) a combination of the overlapping portions **224** and the empty portions **226**. The voids **118** through the material **116** are defined by (i.e., match) whichever of (i) and (ii) from the previous sentence does not define the material **116** within the **114**. The lithography mask **112** is then incorporated into a workpiece **120** that includes the substrate **12**, lithography ink **122** disposed on the substrate **12**, and the lithography mask **112** disposed on the lithography ink **122**.

(103) At a step **230**, the method **200** further includes exposing the workpiece **120** to a curing agent **126**. The curing agent **126**, such as ultraviolet light, transmits through the voids **118** through the material **116** of the lithography mask **112** to cure exposed portions of the lithography ink **122**. The material **116** of the lithography mask **112** blocks non-exposed portions of the lithography ink **122** from exposure to the curing agent **126** and thus the non-exposed portions are not cured. The non-exposed portions are removed from the substrate **12** along with the lithography mask **112**. The exposed portions of the lithography ink **122** remain on the substrate **12** as an etching mask **128**.

(104) At a step **232**, the method **200** further includes contacting the substrate **12** with the etching

mask **128** with an etchant **132**. The etchant **132** selectively etches the etching mask **128**. Depending on how the lithographic mask **112** was formed, the etching mask **128** could allow the etchant **132** to etch into the substrate **12** the free portions **222** of the first ellipses **204** and the second ellipses **210** that do not overlap, and prevent the etchant **132** from etching into the substrate **12** the combination of the overlapping portions **224** and the empty portions **226**. Alternatively, the etching mask **128** could allow the etchant **132** to etch into the substrate **12** the combination of the overlapping portions **224** and the empty portions **226**, and prevent the etchant **132** from etching into the substrate **12** the free portions **222** of the first ellipses **204** and the second ellipses **210** that do not overlap. In any event, after the etching step **232**, the etching mask **128** is removed. The substrate **12** has the textured region **20** on the primary surface **18**.

(105) In embodiments, the method **100** and method **200** each include, after the aforementioned etching steps have been performed, forming the secondary surface features **82** into the one or more sections **80** of the textured region **20**. This step increases the surface roughness (R.sub.a) at the one or more sections **34** to within the range of 5 nm to 100 nm. In embodiments, the step of forming the secondary surface features **82** into one or more sections **80** of the textured region **20** comprises contacting the one or more sections **80** of the textured region **20** of the substrate **12** with another etchant. This etchant is different than the etchant **132**. In embodiments, this etchant includes acetic acid and ammonium fluoride. In embodiments, the etchant includes (in wt %): 85 to 98 acetic acid, 0.5 to 7.5 ammonium fluoride, and 0 to 11 water. The water can be deionized water. In embodiments, the etchant contacts the one or more sections **80** for a time period within a range of 15 seconds to 5 minutes. In embodiments, the etchant contacts the one or more sections **80** while the etching mask **128** remains on the substrate **12**. After the period of time has concluded the substrate **12** is rinsed with deionized water and dried. The etching steps **130**, **232** and the etching step to form the secondary surface features **82** can be conducted at room temperature.

EXAMPLES

(106) Comparative Example 1A and Example 1B—For Comparative Example 1A, a one-step etching process was utilized to generate a textured region of a substrate with surface features having a circular rather than elliptical perimeter. The surface features were still randomly distributed but had a constant diameter and the perimeters were circular. This is illustrated at FIG. **15A**. The textured surface was illuminated. An image of the reflected light from the textured region was captured. The image is produced at FIG. **15B**. The intensity of the light that the textured surface reflected as a function of scattering angle was measured by using the TSW CASI™ Scatterometer made by the Scatter Works Inc (for light having a wavelength of 550 nm). A graph of the measurements is reproduced at FIG. **15C**.

(107) As the image of FIG. **15B** reveals, surface features having a circular perimeter, although randomly distributed, generate color artifacts when reflecting light. Color labels have been added to the FIG. **15B**, in case the graph is reproduced in black and white. The reader should understand that the reflected light resembles circular rainbows that repeat moving outward from the center. The circular perimeter of the surface features causes the circular color rings.

(108) The graph of FIG. **15C** also illustrates that aspect. The graph is for the specific wavelength of 550 nm, which appears greenish yellow. Intensity of the reflected light at this wavelength is shown to peak several times as a function of scattering angle. The scattering angle at which peak intensity occurs is additionally a function of wavelength. Thus, the scattering angles at which intensity peaks occur for the wavelength of 550 nm will be different than the scattering angles at which intensity peaks occur for other wavelengths. The wavelength dependency increases as the scattering angle increases as well. This accounts for the color splitting being more obvious in the image of FIG. **15B** moving away from the center of the image. Scattering angle increases as a function of increasing position away from the center of the image.

(109) For Example 1B, a one-step etching process was utilized to generate a textured region of a substrate with surface features having an elliptical rather than a circular perimeter. The surface

features were randomly distributed. This is illustrated at FIG. 16A. The textured surface was illuminated. An image of the reflected light from the textured region was captured. The image is produced at FIG. 16B. The intensity of the light that the textured surface reflected as a function of scattering angle was measured (for light having a wavelength of 550 nm). A graph of the measurements is reproduced at FIG. 16C.

(110) As the image of FIG. 16B reveals, the elliptical surface features of Example 1A generate much less color artifacts upon reflecting ambient light than the circular surface features of Example 1B. The image of FIG. 16B lacks any noticeable color-separated rings.

(111) As the graph of FIG. 16C reveals, the major peak of reflected light intensity occurs at a lower scattering angle (i.e., closer to the center of image of FIG. 16B) than compared to FIG. 15C, and the peak is narrower than the peak in FIG. 15C. Further, the graph of FIG. 16C lacks the second and third peaks as the scattering angle increases. Thus, the elliptical surface features of Example 1B do not generate the same color-separated rings moving outward from the center of the reflection as the circular surface features of Comparative Example 1A do. When the various wavelengths of visible light all produce an intensity peak at approximately the same scattering angle, the reflected light is white (the combination of all the wavelengths) and does not appear separated by color.

(112) Example 2A and Comparative Example 2B—For Example 2A, and in reference to FIGS. 10A, 10B, 17A, and 17B, first random distribution of points in a first area was generated having a minimum point-to-point distance of 13 μm . The points were triangulated and ellipses drawn and scaled down according to the process discussed above. A second random distribution of points in a second area was generated, but this time having a minimum point-to-point distance of 15 μm . The points were again triangulated and ellipses drawn and scaled down. The two areas were superimposed into one area. Using a computer model, the superimposed area was assumed to be a textured region with two elevations. The first elevation, illustrated in white, is where the superimposed ellipses do not overlap. The second elevation, illustrated in black, is where no ellipses are present and where the superimposed ellipses overlap.

(113) The intensity of light (having the 550 nm wavelength) reflecting off the textured region as a function of the angle at which the light is scattered was modeled. A graph of the modeled results is reproduced at FIG. 17A. The peak intensity was at 0.3 degrees, which is desirable to tightly pack the peak intensities of reflected wavelengths to avoid color separation. The graph additionally shows a peak in intensity at about 3.5 degrees. The further away from center (i.e., the higher the number of degrees) these secondary peak intensities occur, the less visible the color separation is. In this instance, at higher saturations, the color separation might be visible. That aspect is illustrated in the computer-modeled images reproduced at FIG. 17B (top), which shows predicted reflection from the textured region as a function of saturation.

(114) For Comparative Example 2B, the same process was used with the same parameters but, instead of ellipses, circles were utilized. Computer-modeled images of predicted reflection from the textured region as a function of saturation are reproduced at FIG. 17B (bottom). As the computer-modeled images show, the surface features generated from overlapping circular segments reflected light having easily distinguishable color bands even at 0% saturation. The color artifacts that the surface features generated from overlapping elliptical segments were not noticeable at saturations lower than 15%, and even at saturations of 15% and 20%, color banding is not readily evident.

(115) Examples 3-5—Examples 3-5 are each a summary of samples where the textured region most resembles that of FIG. 2. The random but specific distribution of the surface features with an elliptical perimeter began with a random distribution of points having a minimum point-to-point distance of 110 μm for Example 3, 25 μm for Example 4, and 10 μm for Example 5. Triangulation was performed and ellipses drawn and scaled so that the ellipses had a fill-fraction of 50%. A one-step etching process produced surface features with an elliptical perimeter at one elevation and a surrounding portion at another elevation from a base-plane. The one-step etching process removed, depending on the sample, 0.11 μm to 0.16 μm of the thickness of the substrate for Example 3, 0.12

μm to 0.17 μm for Example 4, and 0.08 μm to 0.17 μm for Example 5. Various anti-glare performance metrics were measured for all samples. More specifically, the transmission haze (“haze”), the pixel-power deviation (“PPD”), the distinctiveness-of-image (“c-DOI”), specular reflectance (“c-Rspec”), and corrected color shifts ΔC.sub.x_corrected and ΔC.sub.y_corrected were measured and recorded. The results are set forth in Table 1 below. “Min Rspec(λ)” and “max Rspec(λ)” in the tables below refer to far field first surface specular reflectance spectra as measured with a camera with a 2 degree aperture illuminating the sample at a 6 degree angle of incidence. These values are different from the optically coupled Rspec (peak gloss) measurement made with a Rhopoint goniometer (“GU” means Gloss Units). The minimum and maximum are over the visible wavelength range, 400-700 nm.

(116) TABLE-US-00001

TABLE 1	Example	Haze (%)	PPD (%)	c-DOI (%)	c-Rpec (GU)	3	0.6-1.1
	3.0-4.5	19-80	17-36	4	2.7-4.9	1.2-1.6	6-16
							6-16
							5
							1.8-12
							1.4-1.9
							10-64
							10-64
							min
							min
							Rspec(λ)
							Rspec(λ)
							Example (%)
							(%)
							ΔC.sub.x.sub.—.sub.corrected
							ΔC.sub.y.sub.—.sub.corrected
							3
							NA
							NA
							0.10-0.15
							0.04-0.5
							4
							1.63-2.75
							1.93-3.47
							0.51-0.55
							0.49-0.55
							5
							1.14-1.70
							1.88-3.35
							NA
							NA

As mentioned above, Example 3 with the larger surface features produced little measured reflected color artifacts and low transmission haze but had generated a relatively high pixel power deviation. The smaller surface features generated less pixel power deviation but at the expense of increased haze and reflected color artifacts.

(117) Examples 6-9—These examples are each a summary of samples where the textured region most resembles that of FIG. 6, with larger surface features having an elliptical perimeter introduced in a first etching step and then smaller surface features added in a second etching step. The larger surface features for each example began with a random distribution of points having a minimum point-to-point distance of 110 μm. Triangulation was performed and ellipses drawn and scaled so that the ellipses had a fill-fraction of 50. The first etching step removed, depending on the sample, 0.26 μm to 0.28 μm of the thickness of the substrate for Example 6, 0.13 μm to 0.21 μm for Example 7, 0.13 μm to 0.19 μm for Example 8, and 0.13 μm to 0.17 μm for Example 9. For Example 6, smaller surface features having an elliptical perimeter and a fill-fraction of 50%, generated from random distribution algorithm using a minimum point-to-point distance of 25 μm, were added to the textured region in a second etching step. For Example 7, smaller surface features having a circular perimeter, generated from random distribution algorithm using a minimum center-to-center distance of 15 μm to 25 μm, were added to the textured region in a second etching step. For Example 8, smaller surface features having a circular perimeter, generated from random distribution algorithm using a minimum center-to-center distance of 20 μm to 25 μm, were added to the textured region in a second etching step. For Example 9, secondary surface features to impart surface roughness (R.sub.a) within a range of 5 nm to 100 nm were added to the textured region in an etching step. The same anti-glare performance metrics were measured for all samples, and the results are set forth in Table 2 below.

(118) TABLE-US-00002

TABLE 2	Example	Haze (%)	PPD (%)	c-DOI (%)	c-Rpec(GU)	6	3-5
	3.8	24-34	4.6-6.4	7	1.1-2.8	2.4-3.2	53-89
							16-36
							8
							1.3-2.7
							2.0-5.9
							66-84
							19-30
							9
							0.9-3.9
							1.6-4.7
							21-76
							11-20
							min
							min
							Rspec(λ)
							Rspec(λ)
							Example (%)
							(%)
							ΔC.sub.x.sub.—.sub.corrected
							ΔC.sub.y.sub.—.sub.corrected
							6
							1.51-1.63
							2.94-3.18
							NA
							NA
							7
							NA
							NA
							0.28-0.46
							0.11-0.16
							8
							NA
							NA
							0.24-0.28
							0.09-0.11
							9
							1.46-1.56
							1.76-2.14
							0.16-0.19
							0.11-0.15

The incorporation of the smaller surfaces generally resulted in very low reflected color artifacts values. Examples 8 and 9 in particular had measured reflected color artifact values of under 0.3, imperceptible by human eyes.

(119) Examples 10-13—These examples are each a summary of samples where the textured region most resembles that of FIG. 9A, where a first area of larger elliptical perimeters was superimposed with a second area of smaller elliptical perimeters. Then a single etching step produced a textured region with a first elevation including the non-overlapping elliptical segments and a second elevation including the overlapping elliptical segments and areas outside of any elliptical perimeter.

For Examples 11 and 12, the smaller elliptical perimeters that only partially overlapped the larger elliptical perimeters were removed from the design.

(120) The larger elliptical perimeters for each example began with a random distribution of points having a minimum point-to-point distance of 105 μm , except for Example 13, where the minimum point-to-point distance was 15 μm . Triangulation was performed and ellipses drawn and scaled so that the ellipses had a fill-fraction of 50. The minimum point-to-point distances utilized to generate the smaller elliptical perimeters of Examples 10-12 were 25 μm and 13 μm for Example 13.

Triangulation was performed and ellipses drawn and scaled so that the ellipses had fill-fractions of 20 for Examples 10-12 and 50 Example 13.

(121) A single etching step formed the textured region, removing, depending on the sample, 0.12 μm to 0.17 μm of the thickness of the substrate for Example 10, 0.06 μm to 0.20 μm for Example 11, 0.16 μm for Example 12, and 0.14 μm to 0.16 μm for Example 13. With Example 12, secondary surface features to impart surface roughness (R.sub.a) within a range of 5 nm to 100 nm were added to the textured region in an etching step. The same anti-glare performance metrics were measured for all samples, and the results are set forth in Table 3 below.

(122) TABLE-US-00003 TABLE 3 Example Haze (%) PPD (%) c-DOI (%) c-Rpec(GU) 10 2.9-6.0 1.3-2.1 58-90 5.6-18 11 0.4-3.4 1.4-3.1 33-97 11-63 12 3.9-5.7 2.6-2.9 28-35 8-9 13 5.4-7.2 0.9-1.0 96-97 22-27 min min Rspec(λ) Rspec(λ) Example (%) (%) $\Delta C_{\text{sub.x.sub.}} - \text{sub.corrected}$ $\Delta C_{\text{sub.y.sub.}} - \text{sub.corrected}$ 10 1.84-2.55 2.06-3.73 0.13-0.17 0.08-0.10 11 NA NA 0.16 0.16 12 NA NA 0.20-0.48 0.05-0.13 13 0.85-0.98 1.46-3.01 0.00-0.14 0.00-0.16

Example 10 generated very good results, including acceptable haze values, a low pixel power deviation, very low specular reflectance, and reflected color artifacts well under 0.3 and considered to be imperceptible.

(123) Examples 14A-14O—For Examples 14A-14O, a spacing distribution algorithm was utilized to randomly but specifically place points within an area. Each of the points were to be separated by a minimum distance of 105 μm . The points were then triangulated, an inellipse drawn in each triangle, and then the points and triangles were removed. The longest dimension of the ellipses now remaining in the area were scaled down so that the ellipses occupied 50 percent of the area. The placement of the ellipses was then transferred to a lithography mask. The lithography mask was used to form an etching mask on the primary surface of a glass substrate. Each substrate was then etched with the etching mask on the substrate. The etchant utilized had a composition of 0.15 wt % hydrofluoric acid and 1 wt % nitric acid. The etchant contacted the primary surface with the etching mask for a period of time set forth in Table 4 immediately below that varied among the samples. The etchant formed surface features having an elliptical perimeter set into a surrounding portion. The depth of the surface features varied, and the depth for each sample is set forth below.

(124) TABLE-US-00004 TABLE 4 Etching Period Depth of Elliptical Example of Time (seconds) Surface Features (μm) 14A 212 0.186 14B 200 0.179 14C 165 0.1451 14D 150 0.1353 14E.sup. 140 0.1311 14F.sup. 178 0.1639 14G 178 0.1619 14H 167 0.1557 14I.sup. 167 0.1526 14J 133 0.1261 14K 133 0.1261 14L.sup. 150 0.1364 .sup. 14M 178 0.1531 14N 167 0.1436 14O 167 0.1429

(125) After removal of the etching mask, the samples of 14M-14O were then subjected to a second etching step to form secondary surface features at the primary surface. The second etching step used an etchant with a composition of 92 wt % acetic acid, 2 wt % ammonium fluoride, and 6 wt % water (deionized). The etchant contacted the substrate for a time period of 120 seconds. The secondary surface features so formed imparted a surface roughness (R.sub.a) of ~28 nm to the textured region at the primary surface.

(126) The pixel power deviation, distinctness-of-image, specular reflection, and transmission haze were measured for the sample of each of Examples 14A-14O. The measured results are set forth in the graphs of FIGS. 18A-18D, which plot the measured value as a function of the depth of the surface features with the elliptical perimeter. Analysis of the graphs reveal that the secondary

surface features to impart surface roughness of Examples 14M-14O resulted in a lower pixel power deviation and specular reflectance compared to when no such secondary surface features were included in Examples 14A-14L. However, the secondary surface features to impart surface roughness of Examples 14M-14O resulted in a higher distinctness-of-image and transmission haze compared to when no such secondary surface features were included in Examples 14A-14L. In general, the introducing of the secondary surface features to the surface features can be either increase or decrease the distinctness-of-image, which depends on the design of the surface features.

(127) Example 15A-15C—Examples 15A and 15B are two different sets of samples, each with surface features having an elliptical perimeter, just as in Examples 14A-14O. The difference was that for the samples of Example 15A, the etching mask used while forming the surface features was kept on the substrate while the another etching step was performed to generate the secondary surface features. For the samples of Example 15B, the etching mask was removed before the etching step was performed to generate the secondary surface features. Thus, in the samples of Example 15A, the secondary surface features and the added surface roughness were formed only on surfaces provided by the elliptical surface features and not the surrounding portion. In contrast, with the samples of Example 15B, the secondary surface features and the added surface roughness were formed on the entire textured region including both the surrounding portion and the surfaces provided by the elliptical surface features.

(128) A scanning electron microscope captured images of a sample from both Example 15A and Example 15B. The images are reproduced at FIG. 19A. The images on the left show the surface features with the elliptical perimeters set into the surrounding portion. The images in the middle show the secondary surface features. The images on the right show the etching depth of the secondary surface features.

(129) The pixel power deviation, transparency haze, and specular reflectance of samples from both Examples 15A and 15B were measured. A Rhopoint instrument was utilized to determine specular reflectance. The graphs reproduced at FIGS. 19B-19D set forth the measured data. Analysis of the graphs reveal that the samples of Example 15B, where the etching mask was removed before the second etching step to impart secondary surface features throughout the entire textured region, resulted in a lower pixel power deviation but higher transmission haze compared to the samples of Example 15A, where the etching mask was maintained during the second etching step and thus the secondary surface features were imparted only to the surfaces provided by the elliptical surface features.

(130) The Rhopoint instrument utilized to measure specular reflectance did not measure a difference between the samples of Examples 15A and 15B. However, the device could measure differences in specular reflectance when a 6 degree angle of incidence for the light to be reflected and a 2 degree aperture to measure the specular reflectance. The graph reproduced at FIG. 19E shows the measured data for samples of Examples 15A and 15B, as well as for a sample (Example 15C) where only the elliptical surface features were present and did not include the secondary surface features to impart surface roughness. Analysis of the graph of FIG. 19E reveals that the presence of the secondary surface features in Examples 15A and 15B reduced specular reflectance compared to when the secondary surface features were absent in Example 15C. The difference in specular reflectance between Examples 15A and 15B is wavelength dependent.

(131) Example 16—For Example 16, a sample was prepared similar to the samples Examples 14M-14O, where surface features with an elliptical perimeter are set into a surrounding portion in a first etching step forming textured region, and then secondary surface features are etched throughout the entire textured region to increase surface roughness. The sample so prepared was then analyzed with a white light interferometer to measure the three dimensional profile of the textured region. FIG. 20A illustrates the three dimensional profile that was measured. The top half illustrates relative elevation differences between elliptical surface features and the surrounding portion. The bottom half illustrates the topography of the secondary surface features, with the topography of the

secondary surface features added to the surfaces that the elliptical surface features are provided illustrated at the left, and the topography of the secondary surface features added to the surrounding portion illustrated at the right. The three dimensional profile of the secondary features within the elliptical surface features is measurably different than the three dimensional profile of the secondary features at the surrounding portion—with the surrounding portion showing deeper secondary features.

(132) An atomic force microscope was utilized to image and determine the surface roughness (R.sub.a) imparted by the secondary surface features at both (i) a surface provided by an elliptical surface feature and (ii) at the surrounding portion. The images are reproduced at FIG. 20B. The image on the left is of the secondary surface features at the surface provided by the elliptical surface feature, and shows a surface roughness (R.sub.a) of 15.3 nm. The image on the right is of the secondary surface features at the surrounding portion, and shows a surface roughness (R.sub.a) of 33.5 nm. The image on the right and the higher surface roughness (R.sub.a) value at the surrounding portion matches the topography data illustrated at FIG. 20A. The surrounding portion was covered by the etching mask during the formation of the elliptical surface features and thus had not been contacted with an etchant, unlike the elliptical surface features which were created by the first etching step. Thus, it is believed that the surrounding portion, previously untouched by an etchant, was more sensitive to the second etching step to impart the secondary surface features.

Claims

1. A substrate for a display article, the substrate comprising: a glass or glass-ceramic composition; a primary surface; and a textured region on at least a portion of the primary surface, the textured region comprising surface features, each of the surface features comprising a perimeter that is parallel to a base-plane extending through a thickness of the substrate below the textured region, wherein each of the surface features has a predetermined location relative to each other but collectively are arranged with so that no grouping of the surface features is arranged geometrically the same as any other grouping of the surface features.
2. The substrate of claim 1, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher mean elevation from the base-plane and (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation.
3. The substrate of claim 2, wherein the textured region further comprises a surrounding portion, into which the surface features are set, or from which the surface features project; the surrounding portion provides either (i) the one or more higher surfaces or (ii) the one or more lower surfaces; and the surface features provide the other of (i) the one or more higher surfaces and (ii) the one or more lower surfaces, whichever the surrounding portion does not provide.
4. The substrate of claim 2, wherein the higher mean elevation differs from the lower mean elevation by a distance within a range of 0.05 μm to 0.70 μm .
5. The substrate of claim 1, wherein the perimeter of each of the surface features comprises a longest dimension parallel to the base-plane; and the longest dimensions of the perimeters of the surface features are not all parallel to each other.
6. The substrate of claim 5, wherein the longest dimension of the perimeter of each of the surface features is within a range of 5 μm to 150 μm .
7. The substrate of claim 1, wherein the surface features comprise a fill-fraction that is within a range of 40% to 60%.
8. The substrate of claim 1, wherein the surface features comprise larger surface features and smaller surface features; the perimeters of the larger surface features comprise a first range of longest dimensions parallel to the base-plane; the perimeters of the smaller surface features comprise a second range of longest dimensions parallel to the base-plane; the longest dimensions of

the range of longest dimensions of the larger surface features are longer than the longest dimensions of the range of longest dimensions of the smaller surface features; and the first range and the second range are non-overlapping.

9. The substrate of claim 8, wherein the perimeters of the larger surface features totally surround the perimeters of some of the smaller surface features; and the perimeters of some the smaller surface features reside entirely outside of the perimeters.

10. The substrate of claim 8, wherein the perimeter of at least one of the smaller surface features partially overlaps with the perimeter of one of the larger surface features, such that (i) part of the perimeter of the smaller surface features is inside the perimeter of the larger surface feature and (ii) part of the perimeter of the smaller surface features is outside the perimeter of the larger surface feature.

11. The substrate of claim 8, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation, and (iii) a surrounding portion, into which the larger surface features are set, or from which the larger surface features project.

12. The substrate of claim 11, wherein the larger surface features project from the surrounding portion, some of the smaller surface features project from the surrounding portion, and the remaining smaller surface features, those that do not project from the surrounding portion, are set into the larger surface features; the larger surface features and smaller surface features that project from the surrounding portion provide the one or more higher surfaces residing at the higher mean elevation; and the surrounding portion and the smaller surface features that are set into larger surface features provide the one or more lower surfaces residing at the lower mean elevation.

13. The substrate of claim 11, wherein the larger surface features are set into the surrounding portion, some of the smaller surface features are set into the surrounding portion, and the remaining smaller surface features, those that are not set into the surrounding portion, project from the larger surface features; the larger surface features and the smaller surface features that are set into the surrounding portion provide the one or more lower surfaces residing at the lower mean elevation; and the surrounding portion and the smaller surface features that project from the larger surface features provide the one or more higher surfaces residing at the higher mean elevation.

14. The substrate of claim 8, wherein the textured region further comprises (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane that is closer to the base-plane than the higher mean elevation, (iii) one or more intermediate surfaces residing at one or two intermediate mean elevations from the base-plane, wherein the one or two intermediate mean elevations are disposed between the higher mean elevation and the lower mean elevation, and (iv) a surrounding portion, into which the larger surface features are set, or from which the larger surface features project.

15. The substrate of claim 14, wherein the larger surface features provide at least a portion of the one or more higher surfaces residing at the higher mean elevation.

16. The substrate of claim 14, wherein the larger surface features provide at least a portion of the one or more lower surfaces disposed at the lower mean elevation.

17. The substrate of claim 1, wherein the textured region further comprises one or more sections comprising secondary surface features imparting a surface roughness ($R_{\text{sub.a}}$) within a range of 5 nm to 100 nm.

18. The substrate of claim 1, wherein the textured region exhibits a transmission haze within a range of 0.5% to 5.0%; the textured region exhibits a pixel power deviation within a range of 1.0% to 3.0%; the textured region exhibits a distinctness-of-image within a range of 5% to 70%; the textured region exhibits a specular reflectance within a range of 4 GU to 40 GU; and the textured region exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ that are each

respectively within a range of 0.03 to 0.6.

19. The substrate of claim 1, wherein the substrate is a multi-component glass composition, and the surface features have the multi-component glass composition.

20. A substrate for a display article, the substrate comprising: a primary surface; a base-plane extending through the substrate below the primary surface; and a textured region at the primary surface, the textured region comprising (i) one or more higher surfaces residing at a higher mean elevation from the base-plane, (ii) one or more lower surfaces residing at a lower mean elevation from the base-plane, (iii) a first portion providing either the one or more higher surfaces or the one or more lower surfaces, and (iv) a second portion providing the other of the one or more higher surfaces and the one or more lower surfaces, whichever the first portion is not providing; wherein, elliptical perimeters which lie in planes parallel to the base-plane, define the first portion except for overlapping portions of the elliptical perimeters, where the overlapping portions define in part the second portion.

21. The substrate of claim 20, wherein in addition to the overlapping portions of the elliptical perimeters, the second portion of the textured region further comprises any part of the textured region that is not the first portion.

22. The substrate of claim 20, wherein each of the elliptical perimeters comprises a longest dimension parallel to the base-plane; and the longest dimensions of the elliptical perimeters are not all parallel with each other.

23. The substrate of claim 20, wherein the higher mean elevation differs from the second elevation by a distance within a range of 0.02 μm to 0.70 μm .

24. The substrate of claim 20, wherein the elliptical perimeters comprise larger elliptical perimeters and smaller elliptical perimeters, which are smaller than the larger elliptical perimeters; and the first portion of the textured region is bounded by (i) the elliptical perimeters that do not overlap or intersect with the larger elliptical perimeters, (ii) portions of the smaller elliptical perimeters outside of the larger elliptical perimeters that partially overlap with the larger elliptical perimeters, and (iii) portions of the larger elliptical perimeters that do not overlap with the smaller elliptical perimeters.

25. The substrate of claim 24, wherein at least some of the larger elliptical perimeters entirely encompass more than one of the smaller elliptical perimeters.

26. The substrate of claim 24, wherein a fill-fraction of the larger elliptical perimeters is within a range of 40% to 60%; and a fill-fraction of the smaller elliptical perimeters is within a range of 10% to 30%.

27. The substrate of claim 20, wherein the textured region further comprises one or more sections comprising secondary surface features imparting a surface roughness ($R_{\text{sub.a}}$) within a range of 5 nm to 100 nm.

28. The substrate of claim 20, wherein the textured region exhibits a transmission haze within a range of 0.3% to 8.0%; the textured region exhibits a pixel power deviation within a range of 0.7% to 3.5%; the textured region exhibits a distinctness-of-image within a range of 25% to 100%; the textured region exhibits a specular reflectance within a range of 5 GU to 30 GU; and the textured region exhibits corrected color shifts $\Delta C_{\text{sub.x_corrected}}$ and $\Delta C_{\text{sub.y_corrected}}$ that are each respectively within a range of 0.00 to 0.50.

29. The substrate of claim 20, wherein the substrate comprises a glass or glass-ceramic.

30. The substrate of claim 29, wherein the substrate is a multi-component glass composition, and the surface features have the multi-component glass composition.

31. A substrate for a display article, the substrate comprising: a primary surface; and a textured region on at least a portion of the primary surface, the textured region comprising surface features, each of the surface features comprising a perimeter that is parallel to a base-plane extending through a thickness of the substrate below the textured region, wherein each of the surface features has a predetermined location relative to each other but collectively are arranged so that no grouping

of the surface features is arranged geometrically the same as any other grouping of the surface features, wherein, the perimeter of each of the surface features comprises a longest dimension parallel to the base-plane, and wherein, the longest dimensions of the perimeters of the surface features are not all parallel to each other.
